

Decorated logarithmic eta products (OEIS A330499 and A330498)

Uniform oscillatory asymptotics and threshold inverses;
the harmonic Lambert decoration and the $\log 2$ constant of A330498

Two research reports of one external research session
(bundle Reports 185 and 211), merged into one ProveIt report

Part I: 3 October 2026 Part II: 4 October 2026 merged: 7 October 2026

Abstract

Let $L(z) = -\log(1 - z)$, $\rho = 1 - e^{-1}$ and $M = e - 1$. This report prints two manuscripts on the compositions $\sum_{k \geq 1} \log(1 + L(z)^k)$ (OEIS A330499) and $\sum_{k \geq 1} k^{-1} \log(1 + L(z)^k)$ (OEIS A330498), and on the same compositions with any analytic decoration B in place of L that has nonnegative coefficients, a root $B(\rho) = 1$ inside its disk, difference span one and positive variance. Part I proves, by the Dedekind eta transformation and a uniform stationary-phase estimate on the coefficient circle, that $a_n \rho^n / n! = \pi^2 / (12(e - 1)) + \sum_{j < N} n^{-3/4-j/2} H_j(\sqrt{n}) + O(n^{-3/4-N/2})$ for every N , with explicit almost periodic H_j ; this identifies the decimal $0.478656\dots$ recorded in A330499. It shows that the error changes sign at the scale $n^{-3/4}$, that root-of-unity cusps exclude a standard Delta-domain at ρ , and gives two-ceiling inverse envelopes at every order. Part II proves, by a signed divisor Voronoi identity, a Schwartz-space renewal interpolation and a uniform Hankel estimate, that $n \rho^n a(n) / n! = \log 2 + \sum_{j < J} n^{-1/4-j/2} H_j(\sqrt{n}) + O(n^{-1/4-J/2})$, with $P_0, \dots, P_3$ explicit; in particular $a(n) \sim \log 2 (n - 1)! \rho^{-n}$, which proves Kotěšovec's conjecture $c = \log 2$ in A330498. It also gives signed subsequences at scale $n^{-1/4}$, eventual positivity, increase and strict log-convexity, and inverse envelopes. The write records that the two leading oscillations have the same frequencies, damping and phase and differ only in their amplitudes, and places the inverses relative to the repository's transseries volume. Effective constants, other spans, vanishing variance, error laws and higher-order structure remain open. The report is unrefereed, and nothing in it is formalized.

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Guide to this report

The two Parts are two manuscripts of the session bundle of Reports 1–243 (arrival commit 60f54ea06), placed in this directory by commit 8622ca7e5 (batch 110, cluster 110-ETA).

Part	Bundle report	Delivered title	Labels
I	185 (base)	Decorated logarithmic eta products and OEIS A330499: Uniform oscillatory asymptotics and threshold inverses (3 October 2026)	dep:eta:
II	211	Harmonic Lambert decoration and OEIS A330498: All fixed order oscillations and threshold inverses (4 October 2026)	dep:h1:

One framework, two kernels. Both Parts study a composition $\sum_{k \geq 1} w_k \log(1 + B(z)^k)$ with an analytic decoration B that has nonnegative coefficients, a positive root $B(\rho) = 1$ inside its disk of analyticity, difference span one and positive variance; both specialize to $B = L = -\log(1 - z)$, where $\rho = 1 - e^{-1}$, $M = e - 1$ and the size law has mean and variance $\mu = v = M$. Part I takes $w_k = 1$: the arithmetic weight is $c_m = \sigma(\text{odd}(m))/m > 0$, the exact transformation is the Dedekind eta identity, the coefficients have a pole-sized main term $\pi^2/(12\mu)$ after multiplication by ρ^n , and the first correction is $n^{-3/4}H_0(\sqrt{n})$; for $B = L$ this is OEIS A330499. Part II takes $w_k = 1/k$: the weight is the signed $h(m)/m$, the transformation is a signed divisor Voronoi identity, $n\rho^n[z^n]$ tends to $\log 2$, and the first correction is $n^{-1/4}H_0(\sqrt{n})$; for $B = L$ this is OEIS A330498, and the $\log 2$ is the constant that the entry states as a conjecture. Part II continues Part I explicitly (its Section 23, “Relation to Report185”), and Part I excludes A330498 from every theorem (Section 11). Neither theorem specializes the other.

[write] 7 October 2026; the write’s observation. The two leading oscillatory functions have the same frequencies, damping and phase, and differ only in their amplitudes. For a decoration B with mean μ and variance v , Part I’s (20) has, with $\alpha_m = 2\pi^2 m$, the frequency $2\sqrt{\alpha_m/\mu} = 2\sqrt{2\pi^2 m/\mu}$, which is twice Part II’s β_m of (87), and the damping $e^{-\alpha_m d} = e^{-\pi^2 m v/\mu^2}$ of Part II’s (89); and $-\cos(x + \pi/4) = \sin(x - \pi/4)$ for every real x , since both equal $(\sin x - \cos x)/\sqrt{2}$. So

$$H_0^I(u) = \sum_{m \geq 1} c_m \frac{(2\pi^2 m/\mu)^{1/4}}{\sqrt{\pi}} e^{-\pi^2 m v/\mu^2} \sin(2\beta_m u - \frac{\pi}{4}),$$

$$H_{B,0}^{II}(u) = \sum_{m \geq 1} d_o(m) \left(\frac{8\mu}{m}\right)^{1/4} e^{-\pi^2 m v/\mu^2} \sin(2\beta_m u - \frac{\pi}{4}),$$

with the arithmetic weights $c_m = \sigma(\text{odd}(m))/m$ and $d_o(m) = \tau(\text{odd}(m))$ and the powers $m^{1/4}$ and $m^{-1/4}$. For $B = L$ both have damping r^m , $r = e^{-\pi^2/M}$, which is why the two first-harmonic certificates (Lemma 7.1 and (120)) start from the same bound $r < 1/300$. Neither manuscript states this comparison; it does not make either theorem a consequence of the other, because the exact transformations and the main terms differ.

Arrangement. The base of the merge is Report 185; its **Report185.tex** was staged as this **article.tex**, and it is printed first, as Part I: it is the earlier manuscript, the one that Report 211 continues, and its kernel has the positive weight with a combinatorial model. Report 211 is printed second, as Part II. The two prove different theorems about different sequences, so nothing is printed twice.

Numbering. Sections are numbered continuously; statements are numbered within sections and equations continuously through the report, as in both manuscripts. Part I keeps Report 185’s Sections 1–12, equations (1)–(71), Tables 1–3 and all its other numbers. Report 211’s Section k is Section $k + 12$ here (Sections 13–23), and its statements move with their sections (Report 211’s

Theorem 1.1 is Theorem 13.1); its equation (k) is equation ($k + 71$) here (its (6) is (77)), and its Tables 1–2 are Tables 4–5. Section 24, which gathers the open questions of both Parts under Vladimir’s standing rule of 4 October 2026, was added in the write.

Notes added in the write are set in small type and tagged **[write]**, with their date (7 October 2026). No delivered statement was changed and no delivered symbol was renamed.

What is proved, and where

Statement	Where
<i>A330499</i> , $\sum_k \log(1 + L^k)$ (<i>Part I</i>) $a_n \rho^n / n! = C + \sum_{j < N} n^{-3/4-j/2} H_j(\sqrt{n}) + O(n^{-3/4-N/2})$, $C = \pi^2 / (12(e - 1))$; H_0, H_1, H_2 explicit	Theorem 1.1, Section 6
Exact Stirling transform, cycles of cycles with odd-divisor colours, $a_{n+1} > na_n$	Lemma 2.1, Proposition 2.2
Transfer for every admissible decoration, leading term	Theorem 3.1
$\pi^2 / (12\mu)$, finite recipe for every H_j	
Signed subsequences at scale $n^{-3/4}$; no refinement	Lemma 7.1, Theorem 7.2
$C + c_1/n + O(n^{-2})$	
Odd-order root-of-unity cusps; no standard Delta-domain at ρ	Proposition 8.1, Corollary 8.2
Two-ceiling threshold envelopes at every fixed order; first oscillatory inverse shift	Theorem 9.1, Proposition 9.2
<i>A330498</i> , $\sum_k k^{-1} \log(1 + L^k)$ (<i>Part II</i>) $n \rho^n a(n) / n! = \log 2 + \sum_{j < J} n^{-1/4-j/2} H_j(\sqrt{n}) + O(n^{-1/4-J/2})$; the conjecture $c = \log 2$ of A330498 is proved ; $P_0, \dots, P_3$ explicit	Theorem 13.1, Section 13.1
Transfer for every admissible decoration, leading term $\log 2$	Theorem 15.1
Signed subsequences at scale $n^{-1/4}$	Section 20
Eventual positivity, increase and strict log-convexity; two-ceiling inverse envelopes	Corollary 21.1, Theorem 21.2
<i>Open</i> Effective constants and cutoffs, other support spans, vanishing variance, joint phases and error laws, growing orders and exponentially improved expansions, other weights (both Parts); a Delta-domain classification, single-ceiling inverses at jumps, oscillation distributions (Part II). See Section 24.	

The finite computations of the two packages (exact integers and rationals, and floating-point diagnostics) check identities, coefficients and transcriptions; no asymptotic statement rests on them.

The OEIS entries

The write read both entries on 7 October 2026. **A330499** [1] (revision #11 of 17 December 2019, the revision Part I cites), by Václav Kotěšovec (16 December 2019): “Expansion of e.g.f. $\text{Sum}_{\{k \geq 1\}} \log(1 + \log(1/(1 - x))^k)$ ”, with a b-file of $n \leq 400$ and the formula field “ $a(n) \sim n! * c / (1 - \exp(-1))^n$, where $c = 0.478656\dots$ ”. Part I identifies $c = \pi^2 / (12(e - 1)) = 0.47865665562074963706\dots$ (Theorem 1.1); the entry’s six digits are its truncation. The entry states no closed form and no conjecture about one. **A330498** [7] (revision #13 of 17 December 2019), also by Kotěšovec (16 December 2019): “Expansion of e.g.f. $\text{Sum}_{\{k \geq 1\}} \log(1 + \log(1/(1 - x))^k) / k$ ”, with a b-file of $n \leq 400$ and the formula field “ $a(n) \sim n! * c / (n * (1 - \exp(-1))^n)$, where $c = 0.6931\dots$, conjecture: $c = \log(2)$.” Part II’s (77), $a(n) \sim \log 2 (n - 1)! \rho^{-n}$, is exactly that statement with $c = \log 2$, so

Kotěšovec’s conjecture is **proved** (Theorem 13.1). Report 211 could not reopen the live entry and tied its wording to the saved revision #13; the live entry read by the write is the same revision. Nothing was submitted to the OEIS. The OEIS terms shipped as `data/185-eta-fixture21.json` and `data/211-harmonic-repro-b330498.txt` (Kotěšovec’s b-file) are OEIS content under CC BY-SA 4.0.

The inverses and the transseries volume

The repository’s transseries volume [14] has a Part 0 apparatus for inverting rapidly growing functions. Its symbols are subscripted “vol” here ($\kappa_{\text{vol}}, d_{\text{vol}}, L_{\text{vol}}, \Phi_{\text{vol}}, \varrho_{\text{vol}}$). Statement by statement:

- Part I’s Lambert initialization (68), the root of $x_0 \log(x_0/(e\rho)) = y$, and Part II’s (128), the same equation with Y for y : *instances* of the factorial core `p0:prop:factorial-core` with $\kappa_{\text{vol}} = 1$, $d_{\text{vol}} = -(1 + \log \rho)$ and target $L_{\text{vol}} = y$ (resp. Y). That proposition gives $v_{\text{vol}} = W_0(y/(e\rho))$ and $X_{\text{vol}} = e\rho e^{v_{\text{vol}}} = y/v_{\text{vol}}$, which is x_0 ; the side condition $\kappa_{\text{vol}} \log X_{\text{vol}} + d_{\text{vol}} = v_{\text{vol}} > 0$ holds for $y > 0$. The $O(1)$ distance to the true root stated in Part II is Part II’s own estimate.
- Part I’s Lagrange reversion (70): as a formal series in ε , an *instance* of the perturbation-operator series `p0:eq:operator-series` of `p0:prop:operator-form`, with the core $\Phi_{\text{vol}} = \mathcal{F}_*$ of (65) and the perturbation $\varrho_{\text{vol}} = E_J$: Part I’s $D_y = (1/\mathcal{F}'_*(x))D_x$ is the volume’s $\mathcal{D}_0$ and $X'(y) = 1/\Phi'_{\text{vol}}(X)$. That the series may be evaluated at $\varepsilon = 1$ for large y is Part I’s own argument; the write did not compare it with the volume’s remainder-transport theorem.
- Part I’s one-step correction (69), its first oscillatory shift (66) (Proposition 9.2), and Part II’s first inverse correction (127): each is the first-order term $-\varrho_{\text{vol}}(X)/\Phi'_{\text{vol}}(X)$ of the same series, about the cores $x \log(x/(e\rho))$, $\mathcal{F}_*$ and $\log \Gamma(b) - b \log \rho + \log \log 2$, with error bounds proved by the Parts themselves. *Analogues*, not instances of a theorem of the volume.
- Part I’s Theorem 9.1 and Part II’s Theorem 21.2: two-ceiling brackets proved directly at the integers from a pointwise error bound for $\log a_n$, without an admissible interpolation of the sequence; *not* instances of `p0:thm:staircase`. When the two ceilings agree they determine the threshold, the same conclusion as item (2) of that theorem, and both Parts’ warnings that a smooth root cannot be rounded by one ceiling at every jump describe the failure modes of its items (2) and (5).

Neither Part claims novelty for the inversion mechanics.

Notation across the two Parts

Common to both Parts with the same meaning: $L(z) = -\log(1 - z)$, $\rho = 1 - e^{-1}$, $M = e - 1$, the decoration B and its law K with mean μ , variance v and cumulants κ_r , $\text{odd}(m)$, $\begin{bmatrix} n \\ k \end{bmatrix}$, u as the argument of H_j , $r = e^{-\pi^2/M}$, and the cutoffs χ , δ . The symbols below change meaning between the Parts, or inside one; each Part fixes its meaning there. No symbol was renamed.

Symbol	Meanings
$a_n, a(n)$	Part I: a_n is A330499. Part II: $a(n)$ is A330498. Part II: a_k are the Hankel coefficients (102). <i>False reading</i> : Part II’s a_k is not a sequence term.

Symbol	Meanings
C, c_m, h, d_o	Part I: $C = \pi^2/(12M)$ the leading constant; $c_m = \sigma(\text{odd}(m))/m$ (6). Part II: $h(m) = \sum_{d m} (-1)^{d-1}$, $d_o(m) = \tau(\text{odd}(m))$; $C_j(\beta)$ Fourier coefficients; $C_J, C_{N,L}$ constants. <i>False reading</i> : Part I's c_m is not Part II's C_j .
H_j, P_j, Q_j	Both: H_j the oscillatory coefficient functions, but at different scales ($n^{-3/4-j/2}$ in Part I after normalizing by $\rho^n/n!$; $n^{-1/4-j/2}$ in Part II after normalizing by $n\rho^n/n!$). Part I: $P_j(\alpha)$ real, with $Q_j = (-i)^j P_j$ (17), a function of the dual parameter α . Part II: $P_j(b)$ real Laurent polynomials in $b = \beta_m$; Q_1, Q_2, Q_3 kernel coefficients (119). <i>False reading</i> : the two Parts' P_1 are different functions of different variables.
K_m, Φ_m	Part I: $K_m = c_m e^{-A_m/2} A_m^{1/4} / \sqrt{\pi}$, $\Phi_m(u) = 2u\sqrt{A_m} + \pi/4$ (42). Part II: $K_m = d_o(m)(8M/m)^{1/4} r^m$, $\Phi_m(u) = 2\beta_m u - \pi/4$ (74). Part II: K a Hankel order; K_0 the Bessel function.
α_m, A_m, β_m	Part I: $\alpha_m = 2\pi^2 m$, $A_m = \alpha_m/M$. Part II: $\beta_m = \pi\sqrt{2m/\mu}$, which is $\sqrt{A_m}$ for $B = L$. Part II: $A_{12}, \dots, A_4$ coefficients of P_3 ; $A_J(t)$ the smooth model (122). Part I: $\alpha = 1/4 + J/2$ is Part II's exponent in Theorem 21.2. [Corrected 7 October 2026, independent check of the write: read "Part II: $\alpha = 1/4 + J/2$ is the exponent in Theorem 21.2 and (124). Part I: α is the dual parameter of $P_j(\alpha)$, $E_\alpha, J_\alpha, R_{N,\alpha}$, with values α_m ." The label "Part I" above is wrong.]
d, q_1, r_2, W, Y, U	Part I: $d = v/(2\mu^2)$; q_1, r_2 coefficients of the germ $W(\theta)$ (14); $Y(w)$ (15); $U = (2M^2 - 1)/12$. Part II: W the Lambert function and $Y = \log(x/(\sqrt{2\pi} \log 2))$ in (128); $U_n = \rho^n [z^n] F_B$; $q_{n,k}$ a recurrence. <i>False reading</i> : Part II's W is not Part I's germ.
E, J, R, S	Part I: $E_\alpha(\theta)$; $E_J(x) = \log(S_J(x)/C)$; $J_\alpha(n)$ the mode integral; J an order; $R_{N,\alpha}, R_2(A)$; S_J the chart sum. Part II: $E_j(y, u)$ in the recipe; $E_J(n)$ the scaled diagnostic error (129); J an order; $J_\beta(s)$ (105); $R(s, y)$, $R_{J,\beta}$; S_k a sum of copies of K .
$s, \lambda, \phi, \psi, \eta$	Part I: $s = \sqrt{\alpha/(\mu n)}$, $\lambda = \sqrt{\alpha n/\mu}$, $\phi(y) = y + y^{-1}$, ψ the digamma function, η a small constant. Part II: $s = n^{-1/2}$, $\phi(\theta) = B(\rho e^{i\theta})$, $\psi(\theta) = \log \phi(\theta)$ and ψ_0 the digamma function, η a cutoff. <i>False reading</i> : Part II's ψ is not the digamma function.
$N, \mathcal{N}, D$	Part I: N an order; $\mathcal{N}(Y)$ the threshold (56); D_J a constant. Part II: $N(x)$ the threshold (125); $D(b) = \psi_0(b) - \log \rho$; $D_r = \gamma^{(r)}/\gamma$.
σ, τ, γ	Part I: $\sigma(m)$ the divisor sum. Part II: τ the divisor count; $\sigma^2 = v/\mu^3$; γ Euler's constant in (99) and the Gaussian density $\gamma(z)$ in Section 19.
t, T, V, G, g	Part I: $t(\theta) = -\log B(\rho e^{i\theta})$; $T_n, T_n^{(J)}$ diagnostics; $G(x)$ pole-subtracted. Part II: t a real variable; $t_J(x)$ the smooth root; $T(s, y)$ the recipe series; V_n the normalized sequence; g_s, g_j, G_j densities.

Provenance and merge decisions

Two manuscripts of the session bundle of Reports 1–243. Their title blocks read “Report185” and “Report211”, with the subtitles above in the author slot; both PDF author fields read “Research report”; neither names an author, a tool or an addressee, neither pins a repository commit, and neither carries “prepared for private review” wording (Report 211’s README and its Section 22 say that “private review notes” are *excluded* from the package). They arrived in commit 60f54ea06 and were placed by 8622ca7e5 (batch 110, cluster 110-ETA).

- **Part I**, bundle Report 185, archive `Decorated_Eta_Oscillatory_Asymptotics_and_Inverses_Source.zip` (687,806 bytes; 38 files; `Report185.tex`, 1,376 lines, 21 pages, dated 3 October 2026); the base. Contributes A330499, the eta-identity transfer, the cusp obstruction, the signed subsequences and the inverse envelopes.
- **Part II**, bundle Report 211, archive `Report211-repro-v1.zip` (661,485 bytes; 18 files in an inner directory `Report211/`; `Report211.tex`, 948 lines, 16 pages, dated 4 October 2026;

the “repro-v1” edition). Contributes A330498, the Voronoi and renewal transfer, the proof of the log 2 conjecture, eventual shape and the inverse envelopes. Report 211 knows Report 185 (Section 23) but does not embed or cite it in its bibliography.

Where the merge had to choose:

1. *Base and order.* Report 185 first, as the base (see the Guide).
2. *Nothing duplicated.* The two transfer theorems are parallel results for different kernels, and each Part’s proofs are needed for its own theorem; both are printed in full.
3. *Front matter, numbering and bibliography.* The two title blocks and abstracts are printed at the heads of their Parts, with each manuscript’s opening scope paragraph; one table of contents replaces two. The two bibliographies are merged; the two entries keyed `oeis` became `oeisA330499` (Part I) and `oeisA330498` (Part II), and the write added the entry for the transseries volume.
4. *Write additions.* This front matter, Section 24, the dated notes and the label prefixes. Everything else is delivered text.

What was checked, and what was not

The report is unrefereed and not formalized. At intake (dossier of batch 110, 6 October 2026) both manuscripts were read and no mathematical error was found. An independent computation from the Stirling formulas reproduced both prefixes (0, 1, 2, 13, 71, 558, ... and 0, 1, 1, 6, 24, 152, 1230, 12646, ...) and found the scaled errors $n^{3/4}(a_n \rho^n / n! - C) = -0.0035, +0.0041, -0.0017, +0.0021$ and $n^{1/4}(n \rho^n a(n) / n! - \log 2) = -0.0070, +0.0075, -0.0026, +0.0042$ at $n = 100, 200, 400, 600$, bounded and changing sign at the claimed scales; after subtracting $n^{-1/4} H_0(\sqrt{n})$ (39 modes) the residual of Part II times $n^{3/4}$ was -0.089 and -0.041 at $n = 400$ and 600 . The delivered suites were rerun on copies (results in the README).

The write (7 October 2026) read both OEIS entries, checked the rational certificates of Lemma 7.1 and (120) (359101/26730899 and 599/89401), the decoration table of Part I (Table 1), the specialization of P_1 in both Parts ((44) from (19), and (79) from (90)), the identities $h(m) = (1 - v)\tau(u)$ and the Voronoi coefficient $-d_o(m)$, and the observation in the Guide; it reran the exact checkers of both Parts and Part II’s two coefficient programs on copies. What was *not* done: the analytic proofs (uniform stationary phase, Schwartz renewal interpolation, the all-harmonic Hankel estimate, the cusp argument) were read and their displayed algebra spot-checked, but they are the manuscripts’ proofs, not an independent verification. Ardehali–Rosengren, Egger–Steiner and the DLMF sections were not re-read by the write. The PDF and ZIP release tools were not run.

Independent check of the write (7 October 2026). An independent adversarial check of the write, made the same day, rebuilt both delivered manuscripts and diffed each against its Part. The write’s changes are exactly those listed under “Provenance and merge decisions”. The `.aux` files confirm the numbering stated in the Guide: Part I’s 97 labels are unchanged, Part II’s 79 move exactly by 12 sections, 71 equations and 3 tables, and the write added 19 labels. The check read the live entries again (A330499 #11, A330498 #13, with the quoted names and formula fields). It recomputed both sequences from the Stirling formulas to $n = 600$; they equal Kotěšovec’s two b-files ($n \leq 400$), and the shipped A330498 b-file equals the live one. Every intake value above was reproduced as printed: the eight scaled errors, and the residuals $-0.089, -0.041$. Also confirmed: C and its truncation 0.478656; $r < 1/300$ and the certificates 359101/26730899 and 599/89401;

Table 1; $h(m) = (1 - v)\tau(u)$ and the coefficient $-d_o(m)$ for $m \leq 400$. Both P_1 specializations were checked by hand, using $\kappa_3 = M(M^2 + 1)$ for $B = L$.

The Guide’s observation was re-derived. Numerically the two forms of H_0 agree to 10^{-50} . Its amplitude ratio is explicit: the m -th mode of $H_{B,0}^{\text{II}}$ is $\tau(\text{odd } m)\sqrt{2\mu m}/\sigma(\text{odd } m)$ times that of H_0^{I} , which is $\sqrt{2M} = 1.8538\dots$ for $m = 1$ and $B = L$. Accordingly, the values of the two H_0 at $\sqrt{n}$ for $n = 100, 200, 400, 600$ stand in the ratios 1.853 to 1.857.

The transseries classification was checked against the volume’s statements: $\kappa_{\text{vol}} = 1$ and $d_{\text{vol}} = -(1 + \log \rho)$ give $X_{\text{vol}} = y/W_0(y/(e\rho))$, and (70) is `p0:eq:operator-series` term by term. The reruns reproduce the recorded outputs: both exact checkers, and `verify_coefficients.py` (PASS). Report 211’s Section 22 and README do say that private review notes are excluded. The 48 shipped files equal delivered blobs.

One error of the write was corrected, by a dated bracket in the notation table: the attribution of $\alpha = 1/4 + J/2$ (row α_m, A_m, β_m). No error was found in either manuscript.

Relation to neighbouring reports and formal projects

No other placed report treats A330499, A330498 or A003713 (repository search, 6–7 October 2026). Part II’s “Other bounded comparisons” names two reports of this collection, `a301746-divisor-weighted-asymptotics` (A301746 and A294363) and `a124380-signed-moment-asymptotics` (A124380); its descriptions of them are accurate, and neither shares a theorem with this report or answers a question there. The transseries volume [14] is related to the inverses as stated above. Placement in the collection confers no formal status: no Lean or Rocq file in the repository treats these sequences, and no statement of this report is formalized.

Part I

Decorated logarithmic eta products and OEIS A330499: uniform oscillatory asymptotics and threshold inverses

Bundle Report 185, archive `Decorated_Eta_Oscillatory_Asymptotics_and_Inverses_Source.zip`, delivered as `Report185.tex` (21 pages), the base of the merge. Delivered title block “Report185 / Decorated logarithmic eta products and OEIS A330499 / Uniform oscillatory asymptotics and threshold inverses / 3 October 2026”. Printed as delivered, with `[write]` notes; section, statement and equation numbers are those of the delivery. The opening paragraph “Scope” is the manuscript’s own.

Abstract of this Part (as delivered). We prove an all-fixed-orders coefficient theorem for the composition $F_B(z) = \sum_{k \geq 1} \log(1 + B(z)^k)$, where B is analytic with nonnegative coefficients and its occupied sizes have difference span one. If $B(\rho) = 1$ strictly inside the disk of analyticity, the leading coefficient is $\pi^2/(12\mu)$ after multiplication by ρ^n , where μ is the mean of the decorating size law. The first correction is $n^{-3/4}H_0(\sqrt{n})$, followed by half-power orders; each H_j is an explicit entire trigonometric series, bounded and uniformly almost periodic on the real axis. Its mode damping is determined by the variance divided by the squared mean. A pole-subtracted boundary integral and a uniform stationary-phase estimate justify summing the actual remainders, even though accumulating root-of-unity cusps prevent standard Delta-domain transfer. For $B(z) = -\log(1 - z)$, we identify the exact leading constant of A330499, give its first three oscillatory terms, prove signed subsequences that exclude a pure $1/n$ refinement, and derive rounding-aware inverse envelopes to every fixed order. Exact finite checks and optional floating-point diagnostics accompany the proof.

Scope. The reusable contribution is the decorated-product coefficient theorem, including its uniform remainder. The eta transformation, stationary phase, labelled supernecklaces and inverse reversion are classical inputs. The bounded source comparison in Section 11 is not a worldwide priority determination. All orders and the decoration are fixed before $n \rightarrow \infty$. The constants in the asymptotic remainder are existence constants; this report does not supply a certified large- n cutoff or a certified numerical inverse algorithm.

1 Main results and their meaning

Let

$$L(z) = -\log(1 - z), \quad F(z) = \sum_{k \geq 1} \log(1 + L(z)^k) = \sum_{n \geq 0} a_n \frac{z^n}{n!}. \quad (1)$$

All logarithms in the defining series are the power-series logarithms at the origin. The sequence $(a_n)_{n \geq 0}$ is OEIS A330499 [1]. Its first values are

$$0, 1, 2, 13, 71, 558, 5344, 60926, 766898, 10759096.$$

The OEIS record inspected on 3 October 2026 gives the leading equivalent $n!c(1 - e^{-1})^{-n}$ with the displayed decimal $c = 0.478656\dots$. Define

$$\rho = 1 - e^{-1}, \quad M = e - 1, \quad C = \frac{\pi^2}{12M} = 0.478656655620749637061291495659\dots \quad (2)$$

The decimal is a numerical evaluation of the exact expression, not an interval certificate.

[write] **7 October 2026.** The live entry A330499 read by the write on 7 October 2026 is still revision #11 (17 December 2019); its formula field gives only “ $c = 0.478656\dots$ ”, which is the truncation of C to six decimals. The exact value $C = \pi^2/(12(e-1))$ is this report’s; the entry records no conjecture about it (see the front matter, “The OEIS entries”). Nothing was submitted to the OEIS.

Theorem 1.1 (A330499 at every fixed order). *There are real entire functions H_j , $j \geq 0$, bounded on the real axis, such that for every fixed integer $N \geq 1$,*

$$a_n = n! \rho^{-n} \left\{ C + \sum_{j=0}^{N-1} n^{-3/4-j/2} H_j(\sqrt{n}) + \mathcal{O}_N(n^{-3/4-N/2}) \right\}. \quad (3)$$

Each H_j and each of its derivatives is given by an absolutely convergent trigonometric series. The first three functions are explicit in Section 6; a finite recipe for every further function is given in Theorem 3.1. In particular,

$$\frac{a_n \rho^n}{n!} = C + n^{-3/4} H_0(\sqrt{n}) + n^{-5/4} H_1(\sqrt{n}) + n^{-7/4} H_2(\sqrt{n}) + \mathcal{O}(n^{-9/4}). \quad (4)$$

The oscillation in (3) is a power-scale effect, not an exponentially small correction. In particular, there is no constant c_1 for which $a_n \rho^n / n! = C + c_1/n + \mathcal{O}(n^{-2})$. Section 7 proves this from an exact rational first-harmonic tail estimate and explicit signed subsequences. A leading equivalent by itself does not reveal this obstruction.

The proof is most useful in a broader setting. For a nonnegative analytic decoration B , the numbers $b_j \rho^j$ at $B(\rho) = 1$ define a probability law. If its mean and variance are μ and v , the m th oscillation has damping

$$\exp\left(-\frac{\pi^2 m v}{\mu^2}\right). \quad (5)$$

This identifies a structural smoothing parameter. A small variance relative to the squared mean makes many modes relevant; it does not justify taking a zero-variance limit of a theorem whose constants depend on the decoration.

Why the proof uses the coefficient boundary. The modular eta identity is exact, but the coefficient boundary approaches its singular point tangentially. The local real-axis expansion loses the oscillatory information. Moreover, singular preimages of odd-order roots of unity accumulate at ρ from nearly vertical directions. Section 8 proves that these are actual nonremovable cusps. The standard Delta-domain hypothesis for singularity analysis is therefore unavailable. We instead subtract the dominant pole, use dominated boundary coefficient extraction, and control every dual eta mode uniformly before summing its error.

2 Exact coefficients and labelled interpretation

For an integer $m \geq 1$, let $\text{odd}(m)$ be its largest odd divisor, and let $\sigma(m) = \sum_{d|m} d$. Put

$$c_m = \frac{\sigma(\text{odd}(m))}{m}. \quad (6)$$

The distinction between c_m and the leading constant C will be used throughout.

Lemma 2.1 (The positive logarithmic product coefficients). *For $|q| < 1$,*

$$F_0(q) := \sum_{k \geq 1} \log(1 + q^k) = \sum_{m \geq 1} c_m q^m. \quad (7)$$

In particular, $0 < c_m \leq 1 + \log m$.

Proof. The defining logarithmic series converges locally normally. Writing $P(q) = \prod_{k \geq 1} (1 - q^k)$ and using its analytic logarithm gives

$$F_0(q) = \log P(q^2) - \log P(q), \quad \log P(q) = - \sum_{m \geq 1} \sigma_{-1}(m) q^m,$$

where $\sigma_{-1}(m) = \sum_{d|m} d^{-1} = \sigma(m)/m$. Consequently the coefficient is

$$\sigma_{-1}(m) - \mathbf{1}_{2|m} \sigma_{-1}(m/2).$$

Writing $m = 2^\nu u$ with u odd shows that this difference equals $\sigma(u)/m$. Positivity follows, while $c_m \leq \sigma_{-1}(m) \leq \sum_{j=1}^m 1/j \leq 1 + \log m$. Alternatively, expanding the individual logarithms gives the exact identity $c_m = \sum_{j|m} (-1)^{j+1}/j$. $\square$

Unsigned Stirling numbers of the first kind are denoted by $\begin{bmatrix} n \\ k \end{bmatrix}$. They satisfy

$$\frac{L(z)^k}{k!} = \sum_{n \geq k} \begin{bmatrix} n \\ k \end{bmatrix} \frac{z^n}{n!}, \quad \begin{bmatrix} n+1 \\ k \end{bmatrix} = n \begin{bmatrix} n \\ k \end{bmatrix} + \begin{bmatrix} n \\ k-1 \end{bmatrix}.$$

Thus (7) yields the finite exact transform

$$a_n = \sum_{k=1}^n \begin{bmatrix} n \\ k \end{bmatrix} (k-1)! \sigma(\text{odd}(k)), \quad a_0 = 0. \quad (8)$$

This is an integer formula with positive weights. It is also an independent way to evaluate the coefficients without summing truncated analytic products numerically.

Proposition 2.2 (Marked cycles of cycles and strict growth). *The number a_n counts labelled cycles of cycles with the following additional choice: if there are k inner cycles, choose an odd divisor d of k and one of d colors. Furthermore,*

$$a_{n+1} > n a_n \quad (n \geq 1), \quad (9)$$

and $(a_n)_{n \geq 0}$ is strictly increasing.

Proof. There are $\begin{bmatrix} n \\ k \end{bmatrix}$ ways to decompose n distinct labels into k nonempty cycles. These k distinguishable cycles can be arranged into an outer directed cycle in $(k-1)!$ ways. The additional finite choice has $\sum_{d|k, d \text{ odd}} d = \sigma(\text{odd}(k))$ possibilities. This gives (8). The underlying unmarked labelled cycle-of-cycles construction is classical supernecklace combinatorics [5, 6]; the divisor-colored version here is the direct interpretation of the displayed weight.

Write $w_k = (k-1)! \sigma(\text{odd}(k)) > 0$. Applying the Stirling recurrence and taking $\begin{bmatrix} n \\ k \end{bmatrix} = 0$ outside $0 \leq k \leq n$ gives

$$a_{n+1} = n a_n + \sum_{j=0}^n \begin{bmatrix} n \\ j \end{bmatrix} w_{j+1} > n a_n.$$

The last sum is positive for $n \geq 1$. Together with $a_0 = 0$ and $a_1 = 1$, this proves the assertion. $\square$

Unmarked supernecklaces have generating function $-\log(1 - L(z))$ and a different leading scale $n! \rho^{-n}/n$. The finite marking in Proposition 2.2 is not asymptotically innocuous: the logarithmic eta product has a pole-sized leading term and oscillatory power corrections.

3 The general analytic decoration theorem

Assume that

$$B(z) = \sum_{j \geq 1} b_j z^j, \quad b_j \geq 0, \quad B \text{ holomorphic for } |z| < R, \quad 0 < \rho < R, \quad B(\rho) = 1. \quad (10)$$

The support is required to have *difference span one*:

$$\gcd\{j - k : b_j b_k > 0\} = 1. \quad (11)$$

Here the gcd of the differences is the positive generator of the subgroup they span. In particular the support is not a singleton. This hypothesis is stronger than $\gcd\{j : b_j > 0\} = 1$. For example, $(z + z^3)/2$ satisfies the latter condition, but $B(-1) = -1$ gives another boundary contact.

Define the positive-integer-valued variable K by

$$\Pr\{K = j\} = b_j \rho^j, \quad \mu = \mathbb{E}K, \quad v = \text{Var } K > 0. \quad (12)$$

Since $\rho < R$, this law has a moment generating function analytic near zero. Denote its cumulants by κ_r , so $\kappa_1 = \mu$, $\kappa_2 = v$. Let

$$F_B(z) = F_0(B(z)), \quad \alpha_m = 2\pi^2 m, \quad d = \frac{v}{2\mu^2}, \quad q_1 = \frac{\kappa_3}{6\mu^2} - \frac{v^2}{4\mu^3}. \quad (13)$$

The dual-mode parameter α_m is unrelated to the sequence value a_n .

On a neighborhood of real $\theta = 0$, use the logarithm branch with $-\log B(\rho) = 0$, and define the removable germ

$$W(\theta) = \frac{1}{-\log B(\rho e^{i\theta})} - \frac{i}{\mu\theta}, \quad E_\alpha(\theta) = e^{-\alpha(W(\theta)-d)}. \quad (14)$$

The first terms are $W(\theta) = d + iq_1\theta + \mathcal{O}(\theta^2)$. Let

$$Y(w) = \left(\frac{w + \sqrt{w^2 + 4}}{2} \right)^2, \quad Y(0) = 1. \quad (15)$$

The square root is the analytic branch positive at zero. The identity $Y + Y^{-1} = 2 + w^2$ is exact.

Theorem 3.1 (Uniform transfer at every fixed order). *Under (10)–(11), for every fixed integer $N \geq 1$,*

$$\rho^n [z^n] F_B(z) = \frac{\pi^2}{12\mu} + \sum_{j=0}^{N-1} n^{-3/4-j/2} H_j(\sqrt{n}) + \mathcal{O}_{B,N}(n^{-3/4-N/2}). \quad (16)$$

For $\alpha > 0$, form the finite expression

$$Q_j(\alpha) = \sum_{r+k=j} \frac{(-i)^r}{4^r r! k!} \left(\frac{\alpha}{\mu} \right)^{k/2} \left(\frac{\mu}{\alpha} \right)^{r/2} E_\alpha^{(k)}(0) \cdot \frac{d^{2r}}{dw^{2r}} \{Y(w)^k Y'(w)\} \Big|_{w=0}. \quad (17)$$

There is a real function P_j with $Q_j = (-i)^j P_j$; $P_j(\alpha)$ is $\alpha^{-j/2}$ times a polynomial in α of degree at most $2j$. Then

$$H_j(u) = - \sum_{m \geq 1} c_m e^{-\alpha_m d} \frac{(\alpha_m/\mu)^{1/4}}{\sqrt{\pi}} P_j(\alpha_m) \cdot \cos \left(2u \sqrt{\alpha_m/\mu} + \frac{\pi}{4} + \frac{j\pi}{2} \right). \quad (18)$$

Every such series, and each fixed derivative in u , converges absolutely on the real axis and locally uniformly on the complex plane. The functions H_j are real entire and uniformly almost periodic on the real axis; they and all their derivatives are bounded there.

The adjective “entire” does not mean bounded on the complex plane. Uniform almost periodicity means uniform approximation on the real axis by finite trigonometric sums; it follows immediately from the absolute convergence asserted in the theorem.

The first terms of the finite recipe are particularly simple:

$$P_0(\alpha) = 1, \quad P_1(\alpha) = \frac{q_1 \alpha^{3/2}}{\sqrt{\mu}} + \frac{3\sqrt{\mu}}{16\sqrt{\alpha}}. \quad (19)$$

Thus

$$H_0(u) = - \sum_{m \geq 1} c_m e^{-\alpha_m d} \frac{(\alpha_m/\mu)^{1/4}}{\sqrt{\pi}} \cos \left(2u \sqrt{\alpha_m/\mu} + \frac{\pi}{4} \right), \quad (20)$$

$$H_1(u) = \sum_{m \geq 1} c_m e^{-\alpha_m d} \frac{(\alpha_m/\mu)^{1/4}}{\sqrt{\pi}} P_1(\alpha_m) \sin \left(2u \sqrt{\alpha_m/\mu} + \frac{\pi}{4} \right). \quad (21)$$

In view of $\alpha_m d = \pi^2 m v / \mu^2$, these formulas give (5) without fitting an empirical decay constant.

3.1 Examples of admissible decorations

The theorem applies to ordinary analytic functions as written; their interpretation as labelled classes is optional. Table 1 shows common decorations. Each has strictly positive consecutive coefficients, or support $\{1, 2\}$, hence difference span one. For the set decoration write $\ell = \log 2$. These entries

Decoration $B(z)$	ρ	μ	v	v/μ^2
Lists $z/(1-z)$	$1/2$	2	2	$1/2$
Sets $e^z - 1$	ℓ	2ℓ	$2\ell(1-\ell)$	$(1-\ell)/(2\ell)$
Cycles $-\log(1-z)$	$1 - e^{-1}$	M	M	$1/M$
Two sizes $(z + z^2)/2$	1	$3/2$	$1/4$	$1/9$

Table 1: The variance ratio determines exponential mode damping. Every parameter is exact.

follow from $\mu = \rho B'(\rho)$ and $v = \rho^2 B''(\rho) + \rho B'(\rho) - \mu^2$. A monomial decoration has zero variance and is excluded. The theorem is not uniform in a family approaching a monomial, changing its support span, or moving ρ to the boundary of analyticity.

4 The exact eta identity and boundary extraction

The proof of Theorem 3.1 occupies this section and Section 5. No unproved exchange of a fixed-mode asymptotic expansion with an infinite sum is used.

4.1 The transformed product

The classical modular transformation of the Dedekind eta function implies, for positive real t ,

$$P(e^{-t}) = \sqrt{\frac{2\pi}{t}} \exp \left(\frac{t}{24} - \frac{\pi^2}{6t} \right) P(e^{-4\pi^2/t}). \quad (22)$$

The logarithm and square-root branches extend analytically from $t > 0$ to $\operatorname{Re} t > 0$. See DLMF 23.18 [2] and the explicit product identity (15) of Ardehali–Rosengren [3]. Subtracting the logarithm of (22) from the same identity with t replaced by $2t$ gives

$$F_0(e^{-t}) = \frac{\pi^2}{12t} - \frac{\log 2}{2} + \frac{t}{24} - \sum_{m \geq 1} c_m e^{-2\pi^2 m/t}, \quad \operatorname{Re} t > 0. \quad (23)$$

Indeed, with $Q = e^{-2\pi^2/t}$, the remaining product logarithm is $\log P(Q) - \log P(Q^2) = -F_0(Q)$, and $|Q| < 1$ because $\operatorname{Re}(1/t) > 0$. The identity holds first on the positive real axis and then on the entire right half-plane by holomorphic continuation. This also fixes the logarithmic branch without a possible hidden multiple of $2\pi i$.

On the coefficient circle $z = \rho e^{i\theta}$, the cumulant identity $\log B(\rho e^{i\theta}) = \log \mathbb{E} e^{i\theta K}$ gives

$$t(\theta) := -\log B(\rho e^{i\theta}) = -i\mu\theta + \frac{v}{2}\theta^2 + \frac{i\kappa_3}{6}\theta^3 + \mathcal{O}(\theta^4), \quad (24)$$

$$\frac{1}{t(\theta)} = \frac{i}{\mu\theta} + d + iq_1\theta + \mathcal{O}(\theta^2). \quad (25)$$

In particular W in (14) is analytic. Choose a sufficiently small $\delta > 0$ so that it is holomorphic on a neighborhood of $[-\delta, \delta]$ and

$$\operatorname{Re} W(\theta) \geq d/2 \quad (|\theta| \leq \delta, \theta \in \mathbb{R}). \quad (26)$$

This positive real part provides the summable mode damping on the actual boundary.

4.2 Why span one removes the other boundary contacts

By the triangle inequality, $|B(\rho e^{i\theta})| \leq 1$. Equality is possible exactly when all the unit complex numbers $e^{ij\theta}$ for occupied sizes j are identical. Equivalently, $e^{i(j-k)\theta} = 1$ for every pair of occupied sizes. The span-one hypothesis is therefore exactly what implies

$$|B(\rho e^{i\theta})| < 1 \quad (\theta \not\equiv 0 \pmod{2\pi}). \quad (27)$$

On any compact arc avoiding zero, the upper bound is strictly below one. Since B is analytic beyond the circle, the defining $F_0(B(z))$ series extends holomorphically through that arc. Hence its boundary values are smooth there, with all derivatives bounded on compact arcs.

4.3 Subtracting the pole before taking boundary values

Set $x = z/\rho$. A direct Taylor expansion at $x = 1$ gives

$$\frac{1}{-\log B(\rho x)} = \frac{1}{\mu(1-x)} + \frac{v-\mu}{2\mu^2} + \mathcal{O}(1-x). \quad (28)$$

For $|x| < 1$, $\operatorname{Re}(1/(1-x)) \geq 1/2$. The constant contribution to the lower bound in (28) is $1/(2\mu) + (v-\mu)/(2\mu^2) = d$. Thus, after shrinking the local neighborhood,

$$\operatorname{Re} \frac{1}{-\log B(\rho x)} \geq d/2 \quad (|x| < 1, x \text{ near } 1).$$

The dual-mode sum in (23) is consequently bounded there by $\sum_m c_m e^{-\alpha_m d/2} < \infty$.

It follows that

$$G(x) = F_B(\rho x) - \frac{\pi^2}{12\mu(1-x)} \quad (29)$$

is bounded on the unit disk. Near $x = 1$, this follows from (28), the bounded mode sum and the analytic term $t/24$; away from 1, it follows from (27) and compactness. Boundary values exist and are smooth away from 1, and are bounded near it. Dominated convergence on radial circles therefore gives Fourier coefficient extraction from those boundary values. No assumption of analytic continuation across 1 is made.

Choose a real even C^∞ cutoff χ , supported in $(-\delta, \delta)$ and equal to one on a smaller neighborhood of zero. The analytic pole-subtracted part of $\pi^2/(12t)$ and the remaining elementary terms in (23), multiplied by χ , are smooth functions of θ . All terms outside the cutoff are smooth as well. Repeated integration by parts makes their n th Fourier coefficients $\mathcal{O}_L(n^{-L})$ for every fixed L . We have reduced the coefficient problem to

$$\rho^n[z^n]F_B(z) - \frac{\pi^2}{12\mu} = - \sum_{m \geq 1} c_m J_{\alpha_m}(n) + \mathcal{O}_L(n^{-L}), \quad (30)$$

where

$$J_\alpha(n) = \frac{1}{2\pi} \int_{-\delta}^{\delta} \chi(\theta) e^{-i(n\theta + \alpha/(\mu\theta))} e^{-\alpha W(\theta)} d\theta. \quad (31)$$

The value at the single point $\theta = 0$ is irrelevant. The exchange of the sum with the integral is justified by the summable majorant $c_m e^{-\alpha_m d/2}$ from (26).

5 Uniform stationary phase and the summed remainder

The following estimate is the quantitative step needed for the infinite mode sum. Its uniformity is stronger than a fixed- m stationary-phase expansion.

Lemma 5.1 (Uniform single-mode estimate). *Fix $\alpha_0 > 0$ and an integer $N \geq 1$. There are positive constants η, c, C_N and a nonnegative integer M_N , depending on the fixed decoration, N and α_0 , such that for all sufficiently large n and $\alpha_0 \leq \alpha \leq \eta n$,*

$$J_\alpha(n) = n^{-3/4} e^{-\alpha d} \frac{(\alpha/\mu)^{1/4}}{\sqrt{\pi}} \operatorname{Re} \left\{ e^{-i(2\sqrt{\alpha n/\mu} + \pi/4)} \sum_{j=0}^{N-1} n^{-j/2} Q_j(\alpha) \right\} + R_{N,\alpha}(n), \quad (32)$$

with Q_j as in (17) and

$$|R_{N,\alpha}(n)| \leq C_N n^{-3/4-N/2} (1+\alpha)^{M_N} e^{-c\alpha}. \quad (33)$$

The constants do not depend on n or α in this range.

Proof. Put

$$s = \sqrt{\frac{\alpha}{\mu n}}, \quad \lambda = \sqrt{\frac{\alpha n}{\mu}}, \quad \theta = sy. \quad (34)$$

Then

$$n\theta + \frac{\alpha}{\mu\theta} = \lambda\phi(y), \quad \phi(y) = y + y^{-1}.$$

The phase has exactly two real stationary points, $y = 1$ and $y = -1$, and $\phi''(1) = 2$, $\phi''(-1) = -2$. Choose η small enough that $s \leq s_0$ and fixed neighborhoods of these two points lie in $\chi(sy) = 1$ for all allowed s . Use a fixed partition into those two neighborhoods and the nonstationary complement.

Uniform amplitude bounds. On either fixed stationary neighborhood, every fixed-order derivative in y of $e^{-\alpha W(sy)}$ is bounded by

$$C_j(1 + \alpha)^{D_j} e^{-c\alpha}, \quad (35)$$

uniformly for $0 < s \leq s_0$. This follows from the chain rule, bounded derivatives of W , and (26). Factors of s are bounded by fixed constants. The same estimate holds after an analytic coordinate change with bounded derivatives on the chosen compact neighborhood. Constants may change from line to line.

An explicit stationary-phase remainder bound. Near $y = 1$, set $w = (y - 1)/\sqrt{y}$, so $y = Y(w)$ and $\phi(y) = 2 + w^2$. For a compactly supported smooth amplitude f , the Gaussian integral gives

$$\int_{\mathbb{R}} e^{-i\lambda w^2} f(w) dw = e^{-i\pi/4} \sqrt{\frac{\pi}{\lambda}} \sum_{r=0}^{N-1} \frac{(-i)^r}{4^r r! \lambda^r} f^{(2r)}(0) + \mathcal{R}_N. \quad (36)$$

For completeness, Fourier inversion expresses the left side as the Gaussian factor times the integral of $\widehat{f}(\xi) e^{i\xi^2/(4\lambda)}$. Taylor expansion of the last exponential through degree $N - 1$, using its real-axis remainder bound, gives

$$|\mathcal{R}_N| \leq C_N \lambda^{-N-1/2} \int_{\mathbb{R}} |\xi|^{2N} |\widehat{f}(\xi)| d\xi. \quad (37)$$

The weighted Fourier norm is bounded by a fixed combination of the L^1 norms of f and its derivatives through order $2N + 2$: use the L^1 bound for $|\xi| \leq 1$ and integrate by parts $2N + 2$ times for $|\xi| > 1$. This proves a remainder bound uniform over our amplitudes, by (35). One may justify the Gaussian Fourier interchange first with a factor $e^{-\varepsilon w^2}$ and then let $\varepsilon \downarrow 0$.

The actual amplitude here is a fixed cutoff times

$$e^{-\alpha d} E_\alpha(sY(w)) Y'(w).$$

At zero the cutoff is identically one. In the term indexed by r in (36), Taylor-expand $E_\alpha(sY(w))$ in s through degree $N - r - 1$. The retained terms are

$$\frac{s^k}{k!} E_\alpha^{(k)}(0) Y(w)^k Y'(w), \quad r + k < N.$$

Real Taylor remainder estimates, also after the finitely many needed w -derivatives, give a bound polynomial in α times $e^{-c\alpha} s^{N-r}$. The damping remains valid on the real Taylor segments $0 \leq \xi \leq sY(w)$ because of (26). Multiplying by $s\lambda^{-r-1/2}$ gives exactly $n^{-3/4-N/2}$ times a fixed power of α . The remainder in (37), after the outside factor s , has the same required form.

For the retained terms,

$$s\lambda^{-1/2} = (\alpha/\mu)^{1/4} n^{-3/4}, \quad s^k \lambda^{-r} = n^{-(r+k)/2} (\alpha/\mu)^{k/2} (\mu/\alpha)^{r/2}.$$

These identities give precisely (17). Since the coefficients of B are real, $W(-\theta) = \overline{W(\theta)}$. The negative stationary point supplies the complex conjugate of the positive one. The factor 2 from adding them combines with $1/(2\pi)$ and the Gaussian factor $\sqrt{\pi}$ to give $1/\sqrt{\pi}$ in (32).

The nonstationary complement. There are three ranges, each controlled by ordinary integrations by parts with the phase derivative.

1. On a fixed compact set avoiding $0, 1, -1$, $|\phi'|$ is bounded below. After L integrations the contribution, including the outside factor s , is bounded by $C_L s \lambda^{-L} (1 + \alpha)^{D_L} e^{-c\alpha}$.

2. On $0 < |y| < 1/2$, substitute $u = 1/y$. The phase is again $u + u^{-1}$ and the Jacobian is u^{-2} . The transformed amplitude and every needed derivative have integrable tails as $|u| \rightarrow \infty$. Boundary terms there vanish; at the other endpoint the partition cutoff handles the boundary. The same bound follows.
3. On $|y| > 2$, ϕ' is bounded away from zero and derivatives of $1/\phi'$ decay integrably. For $k \geq 1$,

$$\int \left| \frac{d^k}{dy^k} \{ \chi(sy) e^{-\alpha W(sy)} \} \right| dy \leq C_k s^{k-1} (1 + \alpha)^{D_k} e^{-c\alpha}. \quad (38)$$

This follows by changing variables back to θ and the chain rule. Since $s \leq s_0$, the bounds are uniform. Terms with no derivative on the amplitude instead have an integrably decaying derivative of $1/\phi'$. Thus $L \geq 1$ integrations give the same $C_L s \lambda^{-L}$ bound. The outer smooth cutoff removes endpoint terms at the support boundary.

Take $L = N + 1$. The power of n in $s \lambda^{-L}$ is $n^{-1-N/2}$, smaller than the target $n^{-3/4-N/2}$; powers of α are absorbed in $(1 + \alpha)^{M_N}$, since $\alpha \geq \alpha_0$. This proves the lemma. $\square$

5.1 Summing actual mode errors

Apply Lemma 5.1 to $\alpha_m = 2\pi^2 m$ with $\alpha_m \leq \eta n$. The sum of the actual remainders is bounded by

$$C_N n^{-3/4-N/2} \sum_{m \geq 1} c_m (1 + \alpha_m)^{M_N} e^{-c\alpha_m} = \mathcal{O}_{B,N}(n^{-3/4-N/2}). \quad (39)$$

For $\alpha_m > \eta n$, the integral itself satisfies

$$|J_{\alpha_m}(n)| \leq C e^{-\alpha_m d/2}.$$

The entire large-mode tail is exponentially small in n . Each retained coefficient series also has an exponentially small tail there: its coefficient is an exponential times a fixed power of α_m . Therefore the partial sums of the retained terms may be extended to all $m \geq 1$. Finally, choose the smooth-circle remainder order L in (30) large enough. This proves (16) with a true n -independent remainder constant.

5.2 Reality and convergence of the coefficient recipe

The symmetry $W(-\theta) = \overline{W(\theta)}$ implies that $E_\alpha^{(k)}(0)$ is $(-i)^k$ times a real polynomial in α of degree at most k . The derivatives involving Y in (17) are real. Thus $Q_j = (-i)^j P_j$. The factor $\alpha^{(k-r)/2} = \alpha^{-j/2} \alpha^k$ shows that P_j is $\alpha^{-j/2}$ times a polynomial of degree at most $2j$. Only finitely many cumulants enter at any fixed order.

The real part in (32) is $P_j(\alpha) \cos(2u\sqrt{\alpha/\mu} + \pi/4 + j\pi/2)$, which gives (18) including its signs. Derivatives of any fixed order in u add only powers of $\sqrt{\alpha_m}$. For complex u in a compact set, the trigonometric factors have absolute value at most a fixed multiple of $\exp(2|\operatorname{Im} u|\sqrt{\alpha_m/\mu})$. Exponential decay linear in m dominates this square-root growth and every polynomial. Consequently all asserted convergences, analyticity and real-axis boundedness follow. This completes the proof of Theorem 3.1.

6 Explicit terms for the logarithmic decoration

For $B(z) = L(z)$, the solution $B(\rho) = 1$ is $\rho = 1 - e^{-1} < 1$, strictly within its analytic disk. Its positive coefficients occupy all sizes, so the theorem applies. Using the logarithmic moment generating function $\log L(\rho e^s)$ gives

$$\mu = v = M, \quad \kappa_3 = M^3 + M, \quad \kappa_4 = M^4 + 6M^3 + M. \quad (40)$$

For instance $\mu = \rho/(1 - \rho) = M$; the remaining identities follow by successively applying $z d/dz$ to $\log L(z)$ and evaluating at $z = \rho$. In particular,

$$d = \frac{1}{2M}, \quad q_1 = \frac{M}{6} - \frac{1}{12M}. \quad (41)$$

Define

$$A_m = \frac{2\pi^2 m}{M}, \quad K_m = c_m e^{-A_m/2} \frac{A_m^{1/4}}{\sqrt{\pi}}, \quad \Phi_m(u) = 2u\sqrt{A_m} + \frac{\pi}{4}, \quad U = \frac{2M^2 - 1}{12}. \quad (42)$$

Then

$$H_0(u) = - \sum_{m \geq 1} K_m \cos \Phi_m(u), \quad (43)$$

$$H_1(u) = \sum_{m \geq 1} K_m \frac{(8M^2 - 4)A_m^2 + 9}{48\sqrt{A_m}} \sin \Phi_m(u). \quad (44)$$

These expressions are absolutely convergent, and the tail decays rapidly for this particular decoration.

6.1 The third oscillatory coefficient with its sign

For a general decoration, write

$$W(\theta) = d + i q_1 \theta + r_2 \theta^2 + \mathcal{O}(\theta^3).$$

Reciprocal-series expansion of (24) through the next term yields

$$r_2 = -\frac{v^3}{8\mu^4} + \frac{v\kappa_3}{6\mu^3} - \frac{\kappa_4}{24\mu^2}. \quad (45)$$

The three terms with $r + k = 2$ in (17) give

$$Q_2(\alpha) = -\frac{\alpha^3 q_1^2}{2\mu} - \frac{\alpha^2 r_2}{\mu} - \frac{15\alpha q_1}{16} + \frac{15\mu}{512\alpha}. \quad (46)$$

To check the constants directly,

$$Y'(0) = 1, \quad (YY')''(0) = 15/4, \quad Y^{(5)}(0) = -15/16,$$

while $E'_\alpha(0) = -i\alpha q_1$ and $E''_\alpha(0) = -2\alpha r_2 - \alpha^2 q_1^2$. Note that $Q_2 = -P_2$, rather than $Q_2 = P_2$.

For $B = L$, equations (40) and (45) simplify to $r_2 = -M^2/24 - M/12$. To avoid confusing the scaled variable with the universal recipe, denote $Q_2(MA)$ by $R_2(A)$. Then

$$R_2(A) = -\frac{A^3 U^2}{2} + A^2 \left(\frac{M^3}{24} + \frac{M^2}{12} \right) - \frac{15AU}{16} + \frac{15}{512A}, \quad (47)$$

and the third coefficient is

$$H_2(u) = - \sum_{m \geq 1} K_m R_2(A_m) \cos \Phi_m(u). \quad (48)$$

Equations (43), (44), (48), and Theorem 3.1 prove Theorem 1.1.

6.2 An independent Laguerre route for finite diagnostics

Put $w = 1 - z/\rho$. Direct series inversion gives

$$\frac{1}{-\log L(z)} = \frac{1}{Mw} - \frac{M}{6}w + \frac{M^2}{24}w^2 - \frac{7M^3}{180}w^3 + \mathcal{O}(w^4). \quad (49)$$

Thus, for a fixed dual parameter α , the analytic amplitude multiplying $\exp(-A/w)$, $A = \alpha/M$, begins as

$$e^{-\alpha/[-\log L(z)]} = e^{-A/w} \left\{ 1 + \frac{\alpha M}{6}w + M^2 \left(\frac{\alpha^2}{72} - \frac{\alpha}{24} \right) w^2 + \mathcal{O}_\alpha(w^3) \right\}. \quad (50)$$

The associated-Laguerre generating function implies the exact identity

$$[x^n](1-x)^j e^{-A/(1-x)} = e^{-A} L_n^{(-j-1)}(A). \quad (51)$$

It follows simply by multiplying $\sum_{n \geq 0} L_n^{(\nu)}(A)x^n = (1-x)^{-\nu-1} \exp(-Ax/(1-x))$ by e^{-A} and taking $\nu = -j - 1$. Finite jets in (50) therefore give independently computable Laguerre approximants.

The fixed-positive-argument Laguerre expansions in DLMF 18.15(iv) [4] agree with the first terms above. Their stated compact-argument uniformity does not cover a dual parameter tending to infinity. Accordingly, this is a cross-check and numerical route; it is not used in place of the uniform summation argument of Section 5.

7 A rational tail bound and genuine sign changes

For the logarithmic decoration, put

$$r = e^{-\pi^2/M}, \quad K_1 = e^{-\pi^2/M} \frac{(2\pi^2/M)^{1/4}}{\sqrt{\pi}} > 0.$$

Numerically $r = 0.003202319565807\dots$ and $K_1 = 0.0033262016024504409820\dots$. We next prove the bound needed for sign changes using only rational inequalities, without relying on those decimals.

Lemma 7.1 (First-harmonic dominance). *For K_m defined in (42),*

$$\frac{\sum_{m \geq 2} K_m}{K_1} < \frac{359101}{26730899} < \frac{1}{50}. \quad (52)$$

Proof. An elementary upper bound for e is

$$e < \sum_{k=0}^6 \frac{1}{k!} + \frac{1/7!}{1 - 1/8} = \frac{31967}{11760} < \frac{87}{32}.$$

The geometric bound is valid since successive terms after $1/7!$ have ratios at most $1/8$. Hence $M < 55/32$. Archimedes' classical bound $\pi > 223/71 > 157/50$ gives

$$\frac{\pi^2}{M} > \frac{(157/50)^2}{55/32} = \frac{197192}{34375} > \frac{573}{100}.$$

The finite positive rational sum $\sum_{k=0}^{20} (573/100)^k/k!$ is greater than 300. Therefore $e^{\pi^2/M} > 300$, and $r < 1/300$.

By Lemma 2.1,

$$\frac{K_m}{K_1} = c_m m^{1/4} r^{m-1} \leq (1 + \log m) m^{1/4} r^{m-1} \leq m^2 r^{m-1} \quad (m \geq 2).$$

For the last inequality, $1 + \log m \leq m$ and $m^{1/4} \leq m$ suffice. Summing $\sum_{m \geq 1} m^2 r^{m-1} = (1 + r)/(1 - r)^3$ and using $r < 1/300$ yields

$$\sum_{m \geq 2} \frac{K_m}{K_1} < \frac{1 + 1/300}{(1 - 1/300)^3} - 1 = \frac{359101}{26730899} < \frac{1}{50}.$$

All the final comparisons are rational and are replayed by the exact verification program. $\square$

Theorem 7.2 (Signed subsequences). *For A330499,*

$$\liminf_{n \rightarrow \infty} n^{3/4} \left(\frac{a_n \rho^n}{n!} - C \right) \leq -\frac{49K_1}{50}, \quad (53)$$

$$\limsup_{n \rightarrow \infty} n^{3/4} \left(\frac{a_n \rho^n}{n!} - C \right) \geq \frac{49K_1}{50}. \quad (54)$$

In particular the normalized error is not $o(n^{-3/4})$ and admits no pure integer-power refinement beginning with $C + c_1/n$.

Proof. Set $A_1 = 2\pi^2/M$. For all sufficiently large integers j , choose n_j to be a nearest integer to

$$\left(\frac{2\pi j - \pi/4}{2\sqrt{A_1}} \right)^2.$$

Changing this square by at most $1/2$ changes its square root by $\mathcal{O}(1/j)$. Hence $\cos \Phi_1(\sqrt{n_j}) \rightarrow 1$. Equation (43) and Lemma 7.1 imply $\limsup_j H_0(\sqrt{n_j}) \leq -49K_1/50$. Choosing instead a nearest integer to $((2j+1)\pi - \pi/4)/(2\sqrt{A_1})$ makes the first cosine tend to -1 , giving the opposite bound. These are increasing unbounded subsequences after discarding finitely many initial values. Finally, Theorem 1.1 with $N = 1$ gives

$$n^{3/4} \left(\frac{a_n \rho^n}{n!} - C \right) = H_0(\sqrt{n}) + \mathcal{O}(n^{-1/2}).$$

This proves both comparisons. A refinement $C + c_1/n + \mathcal{O}(n^{-2})$ would make this expression tend to zero, a contradiction. $\square$

The strict tail inequality actually permits strict improvements of (53)–(54). No exact value of either extreme is claimed. Their determination would require controlling the joint phase behavior, not just the dominant harmonic.

8 Actual root of unity cusps and the Delta obstruction

The failure of a usual analytic-continuation hypothesis should not be inferred just from the appearance of an infinite product. The following argument proves genuine singularities directly.

Proposition 8.1 (Odd-order cusps). *Let ζ be a primitive root of unity of odd order h . With the analytic logarithmic branch of F_0 in the unit disk,*

$$\lim_{t \downarrow 0} t F_0(\zeta e^{-t}) = \frac{\pi^2}{12h^2}. \quad (55)$$

In particular each such root is a nonremovable singularity.

Proof. Absolute convergence for $|q| < 1$ permits the Lambert representation

$$\log P(q) = - \sum_{r \geq 1} \frac{q^r}{r(1 - q^r)}.$$

At $q = \zeta e^{-t}$ the terms $r = h\ell$ contribute, after multiplication by t ,

$$- \sum_{\ell \geq 1} \frac{t}{h\ell(e^{h\ell t} - 1)} \longrightarrow - \frac{\pi^2}{6h^2}.$$

Dominated convergence is justified by $t/(e^{h\ell t} - 1) \leq 1/(h\ell)$. For r not divisible by h , the finitely many nontrivial root arguments have radial segments bounded away from 1. Thus $|1 - \zeta^r e^{-rt}| \geq c_h > 0$, uniformly in r, t . The absolute contribution of these terms is at most

$$c_h^{-1} \sum_{r \geq 1} \frac{e^{-rt}}{r} = \mathcal{O}_h(\log(1/t)),$$

which disappears after multiplication by t . Since h is odd, ζ^2 still has order h . Applying the same limit to $q^2 = \zeta^2 e^{-2t}$ yields $t \log P(q^2) \rightarrow -\pi^2/(12h^2)$. Subtracting $t \log P(q)$ proves (55). The order h is fixed before taking the limit; no uniform-in- h estimate has been used. $\square$

Corollary 8.2 (No standard Delta-domain at the positive point). *For any decoration in Theorem 3.1, F_B has no holomorphic continuation to a standard Delta-neighborhood of ρ that excludes only an outward wedge of half-angle less than $\pi/2$.*

Proof. Since $B'(\rho) = \mu/\rho > 0$, let g be its local holomorphic inverse near $B(\rho) = 1$. For odd $h \rightarrow \infty$, put $\zeta_h = e^{2\pi i/h}$ and $z_h = g(\zeta_h)$. Taylor expansion gives

$$z_h = \rho + \frac{\rho}{\mu}(\zeta_h - 1) + \mathcal{O}(|\zeta_h - 1|^2), \quad z_h \rightarrow \rho, \quad \arg(z_h - \rho) \rightarrow \pi/2.$$

The conjugate sequence has limiting argument $-\pi/2$. The local inverse image of the disk $|q| < 1$ is connected near ρ and meets the original coefficient disk. Along the radial curves $g(\zeta_h e^{-t})$, Proposition 8.1 gives a divergence at z_h . This is the same analytic branch as F_B by continuation from their common interior region.

A standard Delta-neighborhood consists locally of points sufficiently close to ρ , except for an outward wedge $|\arg(z - \rho)| \leq \phi$ with some $\phi < \pi/2$. For all sufficiently large h , z_h lies strictly inside the remaining neighborhood. The nearby radial curves also lie there and connect to the interior branch through the local inverse image of $|q| < 1$. A holomorphic continuation to that whole neighborhood would therefore have to be regular at z_h , contradicting the divergence. $\square$

This establishes the obstruction needed to select the proof method. It does not classify all root-of-unity cusps or all possible analytic continuation domains. The boundary proof is independently valid and does not use the obstruction as a substitute for an error estimate.

9 Smooth inverses and pointwise integer envelopes

The coefficient theorem yields arbitrarily accurate fixed-order *smooth charts*. The discrete threshold remains a staircase, so its statement must retain rounding explicitly. Throughout this section the sequence is A330499 and the constants are (2).

9.1 Every fixed order with two ceilings

Define, for $Y > 0$,

$$\mathcal{N}(Y) = \min\{n \geq 1 : a_n \geq Y\}. \quad (56)$$

Proposition 2.2 proves strict monotonicity, and Theorem 1.1 proves unboundedness, so this is well-defined. For a fixed integer $J \geq 0$, put

$$S_J(x) = C + \sum_{j=0}^J x^{-3/4-j/2} H_j(\sqrt{x}), \quad (57)$$

$$\mathcal{L}_J(x) = \log \Gamma(x+1) - x \log \rho + \log S_J(x), \quad (58)$$

$$p_J = \frac{3}{4} + \frac{J+1}{2}. \quad (59)$$

These functions are defined for all sufficiently large real x , since $S_J(x) = C + \mathcal{O}(x^{-3/4}) > 0$.

Theorem 9.1 (Pointwise threshold envelopes). *For each fixed $J \geq 0$, there are $D_J > 0$ and a starting point x_J such that the charts $\mathcal{L}_J(x) \pm D_J x^{-p_J}$ are strictly increasing on $x \geq x_J$, and their large roots defined by*

$$\mathcal{L}_J(x_+) + D_J x_+^{-p_J} = \log Y, \quad (60)$$

$$\mathcal{L}_J(x_-) - D_J x_-^{-p_J} = \log Y \quad (61)$$

satisfy, for every sufficiently large Y ,

$$\lceil x_+ \rceil \leq \mathcal{N}(Y) \leq \lfloor x_- \rfloor. \quad (62)$$

Moreover $x_+ \leq x_-$ and

$$x_- - x_+ = \mathcal{O}_J \left(\frac{x_+^{-p_J}}{\log x_+} \right). \quad (63)$$

The constants and starting point are existence statements, not numerically certified values.

Proof. Theorem 1.1 with $N = J + 1$, followed by the logarithm on a fixed neighborhood of the positive constant C , gives

$$|\log a_n - \mathcal{L}_J(n)| \leq D_J n^{-p_J} \quad (64)$$

for all sufficiently large integers n , after choosing D_J . Boundedness of the derivatives of H_j gives $S'_J(x) = \mathcal{O}_J(x^{-5/4})$. The standard digamma expansion implies

$$\mathcal{L}'_J(x) = \psi(x+1) - \log \rho + \frac{S'_J(x)}{S_J(x)} = \log(x/\rho) + \mathcal{O}(1/x) + \mathcal{O}_J(x^{-5/4}).$$

Both perturbed charts therefore have positive derivative for large x and tend to infinity. Their roots exist uniquely there, and the upper chart reaches a given height first, so $x_+ \leq x_-$.

If an integer $n < x_+$ is sufficiently large, its upper chart is less than $\log Y$, and (64) forces $a_n < Y$. The finitely many smaller integers also satisfy this once Y is large enough. Hence $\mathcal{N}(Y) \geq \lceil x_+ \rceil$. For $n = \lfloor x_- \rfloor$, the lower chart is at least $\log Y$, so $a_n \geq Y$ and the upper bound follows. This reasoning remains valid if either root is an integer, or if Y is arbitrarily close to a sequence value.

Subtracting the root equations and using the lower derivative bound $c \log x$ for $\mathcal{L}_J$ gives (63) by the mean value theorem. The roots are asymptotically equivalent, as follows already from their common leading factorial scale. $\square$

For large Y the two ceilings are equal or adjacent, since the real root gap tends to zero. They cannot uniformly be collapsed to one ceiling of the central chart. Arbitrarily small changes in Y near a jump can change $\mathcal{N}(Y)$, however accurate the smooth chart may be.

[write] **7 October 2026.** Theorem 9.1 brackets the threshold directly at the integers and is not an instance of the transseries volume's `p0:thm:staircase`; the Lambert seed (68) is an instance of its `p0:prop:factorial-core`, and the reversion (70) of its `p0:prop:operator-form` as a formal series. The front matter, “The inverses and the transseries volume”, gives the classification statement by statement.

9.2 The first oscillatory displacement of the exact core

Let $x_* = x_*(Y)$ be the unique sufficiently large root of the *exact factorial core*

$$\mathcal{F}_*(x) = \log \Gamma(x+1) - x \log \rho + \log C = \log Y. \quad (65)$$

Its derivative is $\mathcal{F}'_*(x) = \psi(x+1) - \log \rho \sim \log(x/\rho)$.

Proposition 9.2 (First oscillatory inverse shift). *Set*

$$x_{\text{osc}} = x_* - \frac{x_*^{-3/4} H_0(\sqrt{x_*})}{C\{\psi(x_*+1) - \log \rho\}}. \quad (66)$$

The central smooth root of $\mathcal{L}_0(x) = \log Y$ differs from x_{osc} by $\mathcal{O}(x_*^{-3/2}/\log x_*)$. In particular there is a fixed $K > 0$ such that the actual integer threshold satisfies

$$\left\lceil x_{\text{osc}} - \frac{Kx_*^{-5/4}}{\log x_*} \right\rceil \leq \mathcal{N}(Y) \leq \left\lceil x_{\text{osc}} + \frac{Kx_*^{-5/4}}{\log x_*} \right\rceil \quad (67)$$

for all sufficiently large Y .

Proof. Write $E_0(x) = \log(S_0(x)/C)$. Then

$$E_0(x) = C^{-1}x^{-3/4}H_0(\sqrt{x}) + \mathcal{O}(x^{-3/2}), \quad E'_0(x) = \mathcal{O}(x^{-5/4}).$$

The central root of $\mathcal{F}_* + E_0$ differs from x_* by $\mathcal{O}(x_*^{-3/4}/\log x_*)$. Taylor expansion over that interval shows that its first displacement is $-E_0(x_*)/\mathcal{F}'_*(x_*)$, with an error bounded by $\mathcal{O}(x_*^{-2}/(\log x_*)^2)$ from varying E_0 and smaller core-curvature terms. Replacing $E_0(x_*)$ by its first displayed term leaves the stated $\mathcal{O}(x_*^{-3/2}/\log x_*)$ error. The two roots of Theorem 9.1 for $J = 0$ are within $\mathcal{O}(x_*^{-5/4}/\log x_*)$ of the central root. Enlarge a single constant K to contain all these errors and take ceilings. $\square$

9.3 Lambert initialization and the accuracy it does not supply

Write $y = \log Y$ and use the positive real branch of Lambert W . The solution of $x_0 \log(x_0/(e\rho)) = y$ is

$$x_0 = \frac{y}{W(y/(e\rho))}. \quad (68)$$

Stirling's expansion gives

$$\mathcal{F}_*(x) = x \log \frac{x}{e\rho} + \frac{1}{2} \log(2\pi x) + \log C + \frac{1}{12x} + \mathcal{O}(x^{-3}).$$

Since the omitted half-logarithmic term is $\mathcal{O}(\log x_0)$ and the core derivative is of order $\log x_0$, the exact core root first differs from x_0 by $\mathcal{O}(1)$. One Taylor correction yields

$$x_* = x_0 - \frac{\frac{1}{2} \log(2\pi x_0) + \log C}{\log(x_0/\rho)} + \mathcal{O}\left(\frac{1}{x_0 \log x_0}\right). \quad (69)$$

Indeed the remaining Stirling term, derivative variation and quadratic core term are all $\mathcal{O}(1/x_0)$ in the equation, hence $\mathcal{O}(1/(x_0 \log x_0))$ in the root.

This Lambert expression is a useful seed for solving (65). Its displayed error is smaller than the leading oscillatory displacement, but larger than the next half-power inverse corrections. Therefore replacing x_* in an arbitrarily high-order claim by this finite seed without propagating its error would be incorrect. The root in (66) is the exact factorial-core root, not just (69).

9.4 Generating higher smooth inverse terms

For any fixed J , put $E_J(x) = \log(S_J(x)/C)$, so that $\mathcal{L}_J = \mathcal{F}_* + E_J$. Let $X(y)$ be the large local inverse of $\mathcal{F}_*$ and write $D_y = (1/\mathcal{F}'_*(x))D_x$ when acting on a function of $x = X(y)$. Ordinary Lagrange reversion of the perturbation parameter ε in $\mathcal{F}_*(x) + \varepsilon E_J(x) = y$ gives

$$x = X(y) + \sum_{k \geq 1} \frac{(-\varepsilon)^k}{k!} D_y^{k-1} \left\{ X'(y) E_J(X(y))^k \right\}. \quad (70)$$

This is a local analytic identity in ε near zero and a constructive finite-order reversion rule. For sufficiently large y , one can set $\varepsilon = 1$: in a fixed complex neighborhood of the real root the perturbation and its first derivative tend to zero, whereas the core derivative has size $\log X(y)$, so the analytic implicit solution exists on a parameter disk containing $|\varepsilon| \leq 1$. Equivalently, finite Taylor reversion with a remainder follows by repeated substitution and the same derivative bounds.

Only finitely many derivatives and products are needed for any requested fixed precision. Increasing J reduces the coefficient backward error; reversion then converts it into a smooth root error of order $x^{-p_J}/\log x$. Finally, Theorem 9.1 converts that information to a pointwise integer statement. These are classical inverse-chart and backward-error operations; the model-specific input is the oscillatory coefficient law. They do not create an unambiguous real inverse of the discrete staircase.

10 Computational checks and reproducibility

The companion archive separates mathematical proof, exact finite checks and floating-point diagnostics. Agreement at many indices is useful for detecting algebraic errors, but is not a replacement for Lemma 5.1, nor an interval certificate for its implicit constant.

10.1 Exact mandatory checks

The mandatory Python verification kernel uses only the standard library. It computes the positive Stirling transform (8), matches all 21 terms listed in the inspected OEIS record, checks the direct alternating-divisor identity through 100, checks the exact recurrence and strict growth through 600, and replays the rational inequalities in Lemma 7.1. Independent finite coefficient computations test the formal composition directly rather than reusing the same weighted transform.

Checks use explicit exception-raising guards, so Python's optimization flag cannot silently disable them. Negative controls alter fixtures and other mathematical inputs and must fail in both ordinary

Python and Python -0. A fixed checksum of the exact integer output through 600 is a reproducibility fingerprint; it is not evidence in place of the underlying identities.

10.2 Optional floating diagnostics

The optional high-precision checks use `mpmath`; the FFT and two-size experiment additionally use the numerical dependencies listed in the archive. They are excluded from the standard-library mathematical gate. The portable diagnostic scripts and the historical numerical outputs are retained with provenance and hashes, and the high-precision replay compares 50- and 80-digit computations. Finite tolerances are regression limits on those samples, not uniform asymptotic bounds.

For numerical comparison define

$$T_n = n^{3/4} \left(\frac{a_n \rho^n}{n!} - C \right), \quad T_n^{(J)} = \sum_{j=0}^J n^{-j/2} H_j(\sqrt{n}).$$

The first two error columns in Table 2 illustrate that T_n can remain visibly displaced from $H_0(\sqrt{n})$ at moderate indices. Retaining H_1 generally helps, but asymptotic truncation does not promise monotone improvement at every individual n .

n	T_n	$T_n^{(0)}$	$T_n^{(1)}$
100	-0.003509157374	-0.002844061648	-0.005567577555
200	0.004071822395	0.002461800049	0.004970742765
400	-0.001726451607	0.000985440349	-0.001563360516
800	0.000498855063	0.002121973187	0.000685065706
1000	0.001520405923	-0.000167680657	0.001516496146
2000	0.003100045820	0.002332031756	0.003185266147

Table 2: Finite numerical diagnostics, rounded for display. Exact integer coefficients underlie T_n ; harmonic sums are numerical. No entry is an interval enclosure.

For inverse diagnostics one may take $Y = a_n$, whose exact threshold is n , and compare smooth roots with that known integer. Table 3 uses the exact factorial core, the first shift (66), and the root of the three-term chart $\mathcal{L}_2$. The small residuals illustrate the formulas only; they do not license one-ceiling rounding near every threshold.

n	$x_* - n$	$x_{\text{osc}} - n$	$\mathcal{L}_2^{-1}(\log a_n) - n$
100	-4.57425935 10^{-5}	-8.67428957 10^{-6}	1.05433425 10^{-6}
200	2.77698434 10^{-5}	1.09789162 10^{-5}	-1.46614723 10^{-7}
400	-6.25087523 10^{-6}	-9.81874913 10^{-6}	6.74257110 10^{-8}
800	9.69829090 10^{-7}	-3.15553380 10^{-6}	4.07413144 10^{-9}
1000	2.42462740 10^{-6}	2.69203273 10^{-6}	-7.70215372 10^{-9}
2000	2.68683539 10^{-6}	6.65621847 10^{-7}	-1.17990530 10^{-9}

Table 3: Floating-point smooth inverse errors at known jump values. A first correction need not improve every finite case.

Two independent diagnostics are especially useful for detecting model-specific mistakes. The two-size decoration has the finite coefficient formula

$$[z^n]F_{(z+z^2)/2}(z) = \sum_{k=\lceil n/2 \rceil}^n c_k 2^{-k} \binom{k}{n-k}. \quad (71)$$

Here $\rho = 1$, $\mu = 3/2$, $v = 1/4$, $\kappa_3 = 0$ and $q_1 = -1/216$. Comparisons through $n = 20000$ test the universal first correction against a coefficient formula unrelated to Stirling numbers. Separately, a modular-transform FFT with 262144 samples and 80 eta modes extracts coefficients through selected indices up to 64000, with overlap checks against exact integers through 2000. These FFT outputs are ordinary floating-point experiments, not validated quadrature or interval calculations.

10.3 Rebuilding and the integrity boundary

The archive contains the editable $\text{\LaTeX}$ source, the PDF, the exact kernel, optional experiments, machine-readable check results, a manifest and rebuilding instructions. The build uses a fresh working directory, rejects relevant $\text{\LaTeX}$ warnings and layout overflows, and writes a deterministically ordered ZIP with normalized metadata. The release is checked by clean extraction and rebuilding; changing a hashed payload must be rejected by its integrity checker.

These guarantees concern the specified package and toolchain. Different $\text{\TeX}$ or font versions may change PDF bytes while preserving mathematical content. Full analytic proofs are in this manuscript; no third-party papers or private research notes are needed for a rebuild. See the archive’s reproducibility guide for the exact commands, optional dependencies and the distinction between the exact and numerical test tiers.

[write] **7 October 2026.** In this report the archive’s programs are `code/185-eta-*.py` (its `optional/` programs as `code/185-eta-optional-*.py`), its certificates, data, generated and historical outputs are `data/185-eta-*`, and its `README_REPRODUCIBILITY.md` and `optional/README.md` are `185-eta-README_REPRODUCIBILITY.md` and `185-eta-optional-README.md`. The delivered PDF and the checksum manifest `SHA256SUMS.json` are not shipped, and the delivered `README.md` is replaced by this report’s; the reproducer, the build and the frozen-source checks need the delivered layout and manifest, so they run only in a re-extracted archive. The report’s `README` gives a route that runs the exact kernel on a copy.

11 Source comparison and attribution

The public sources below were reviewed for the mathematical model and proof inputs. The comparison is deliberately bounded.

OEIS model. The official A330499 record [1], revision 11 dated 17 December 2019, defines (1) and lists the leading numerical constant. Its inspected formula field does not give the closed form (2), a proof, or higher corrections. The adjacent A330498 has an additional harmonic $1/k$ weight in its kernel. It does not satisfy the eta identity used here and is excluded from every theorem in this report.

[write] **7 October 2026.** In this merged report “this report” is Part I. Part II (Report 211) treats A330498 with a signed Voronoi identity in place of the eta identity, and proves the constant $\log 2$ that the entry states as a conjecture (Theorem 13.1).

Eta transformation. The modular identity is classical, as recorded in DLMF 23.18 [2]. Ardehali–Rosengren’s 2026 paper [3] provides an explicit product formulation in equation (15). Its small-parameter asymptotic discussion fixes a nontangential angular range in the right half-plane. The present boundary $t(\theta) = -i\mu\theta + v\theta^2/2 + \dots$ is tangential. That source’s displayed radial-type expansions therefore do not, by themselves, prove the decorated coefficient theorem. Our use of the exact identity is followed by a separate uniform boundary argument.

Classical coefficient and inverse machinery. The supernecklace construction is recorded in OEIS A003713 and the Flajolet–Sedgewick book companion, exercise IV.28 [5, 6]. DLMF 18.15(iv) [4] supplies classical Laguerre asymptotics, with its stated compact-positive-argument uniformity. The stationary-phase Gaussian expansion, Lambert factorial initialization, local Lagrange reversion and backward-error conversion are standard methods. The specific contribution here is their uniform, summable application to a nonmonomial analytic decoration and the ensuing explicit oscillatory laws and threshold envelopes.

Bounded overlap screen. Exact-sequence and broader model searches in the available report collection and mathematical repository found no inspected report with this A330499 decorated-logarithmic-product theorem. Potentially adjacent inspected models involved critical polylogarithmic Lambert charts, divisor-weighted exponentiated products, planar-map coefficients on a different logarithmic scale, or factorially forced recurrences. They did not supply the present boundary transfer. This records the scope of the comparison, not proof of complete private-collection coverage or worldwide novelty. No external repository or OEIS record is changed by this report.

12 Further questions and present limits

The theorem gives every fixed order for a fixed decoration. Several natural extensions require additional work.

1. **Effective error constants and cutoffs.** The proof tracks polynomial-times-exponential bounds but does not optimize explicit cutoff radii, amplitude norms or stationary-phase constants. An interval implementation of these choices could turn Theorem 9.1 into a certified inverse algorithm. The current finite diagnostics do not supply those missing constants.
2. **Other support spans.** If the difference span exceeds one, multiple unit-modulus boundary contacts occur. Their phases and cusp types must be combined. Merely replacing the span-one condition by gcd of the occupied sizes is invalid, as $(z + z^3)/2$ demonstrates.
3. **Vanishing variance.** As v/μ^2 decreases, the number of appreciable harmonics grows. A uniform crossover theorem toward monomial decorations would need new two-parameter estimates. It is not a consequence of keeping B fixed in Theorem 3.1.
4. **Joint phases and limiting error laws.** The frequencies are proportional to $\sqrt{m}$. Their arithmetic relations may permit sharper extreme-value or distribution results for $H_0(\sqrt{n})$. The signed subsequences here need only the first harmonic and establish neither an empirical limit law nor the exact limsup and liminf.
5. **Growing orders and exponentially improved expansions.** The coefficient recipe is finite at every fixed order. It does not control coefficient growth as the requested order tends to infinity, optimal truncation, or a complete exponentially improved transseries.
6. **Weighted kernels.** The harmonic weighting in A330498 leads to a different transformation problem. Its neighboring sequence number is not evidence that the same theorem applies. Other products or weights should be treated from their own exact kernels.

[write] **7 October 2026.** These six questions, with the limits stated after Theorem 7.2 and Corollary 8.2, are gathered with Part II's in Section 24. Item 6 is answered for the weight $1/k$ by Part II; other weights remain open.

Conclusion. For A330499, the exact leading constant is only the first part of the answer. Analytic decoration damps the eta modes by a variance ratio but leaves a real $n^{-3/4}$ oscillation, followed by explicitly computable half-power orders. A uniform boundary proof makes that infinite oscillatory expansion rigorous and gives inverse envelopes that remain valid at the jumps of the integer threshold.

Part II

Harmonic Lambert decoration and OEIS A330498: all fixed order oscillations and threshold inverses

Bundle Report 211, archive `Report211-repro-v1.zip`, delivered as `Report211/Report211.tex` (16 pages). Delivered title block “Report211 / Harmonic Lambert decoration and OEIS A330498 / All fixed order oscillations and threshold inverses / 4 October 2026”. Printed as delivered, with `[write]` notes; Report 211’s Section k is Section $k+12$ here and its statements move with their sections; its equation (k) is equation $(k+71)$ here, and its Tables 1–2 are Tables 4–5. The opening paragraph “Scope” is the manuscript’s own.

Abstract of this Part (as delivered). For the exponential generating function $\sum_{k \geq 1} k^{-1} \log(1 + [-\log(1 - z)]^k)$, we prove the $\log 2$ leading constant labelled conjectural in the inspected OEIS A330498 record. After normalization, its first error is a nonvanishing, sign-changing oscillation of order $n^{-1/4}$; every successive fixed order advances by $n^{-1/2}$. We prove a transfer theorem for any fixed analytic decoration with nonnegative coefficients, positive variance and difference span one. A signed divisor Voronoi identity, a Schwartz-space renewal interpolation, and a uniform all-harmonic Hankel estimate give the expansion with its remainder. We display P_0 through P_3 , provide a finite coefficient recipe, and deduce eventual positivity, increase, strict log-convexity, and shrinking two-ceiling inverse envelopes. Exact arithmetic and portable high-precision computations accompany the proof as finite checks.

Scope. The decoration and expansion order are fixed before $n \rightarrow \infty$. Remainder constants and starting indices are existential, not certified numerical error bars. Classical Voronoi summation, Hankel asymptotics, and Fourier local-limit and renewal ideas are credited. The source screen is bounded: this report makes no worldwide first-proof claim, no optimal-truncation claim, and no global transseries claim.

13 Main result and the conjectural constant

Let

$$L(z) = -\log(1 - z), \quad F(z) = \sum_{k \geq 1} \frac{\log(1 + L(z)^k)}{k} = \sum_{n \geq 0} a(n) \frac{z^n}{n!}. \quad (72)$$

All logarithms in this definition are power-series logarithms at zero. This is OEIS A330498. The inspected text export, revision 13 dated 17 December 2019, gives the equivalent $a(n) \sim c(n-1)!(1 - e^{-1})^{-n}$ and labels $c = \log 2$ conjectural [7]. This is a statement about that inspected record, not a claim that every later proof or related paper has been excluded.

`[write]` **7 October 2026.** The write read the live entry on 7 October 2026: it is still revision #13 (17 December 2019), by Václav Kotěšovec (16 December 2019), with the formula field “ $a(n) \sim n! * c / (n * (1 - \exp(-1))^n)$ ”, where $c = 0.6931\dots$, conjecture: $c = \log(2)$.” Equation (77) is exactly that statement with $c = \log 2$, so the theorem proves the conjecture as stated. Nothing was submitted to the OEIS.

Set

$$\rho = 1 - e^{-1}, \quad M = e - 1, \quad \beta_m = \pi \sqrt{\frac{2m}{M}}, \quad r = e^{-\pi^2/M}, \quad d_o(m) = \tau(\text{odd}(m)), \quad (73)$$

where τ is the divisor-counting function and $\text{odd}(m)$ is the odd part of m . Define

$$K_m = d_o(m) \left(\frac{8M}{m} \right)^{1/4} r^m, \quad \Phi_m(u) = 2\beta_m u - \frac{\pi}{4}. \quad (74)$$

Theorem 13.1. *For every fixed integer $J \geq 1$,*

$$\frac{n \rho^n a(n)}{n!} = \log 2 + \sum_{j=0}^{J-1} n^{-1/4-j/2} H_j(\sqrt{n}) + \mathcal{O}_J(n^{-1/4-J/2}), \quad (75)$$

where

$$H_j(u) = \sum_{m \geq 1} K_m P_j(\beta_m) \sin \left(\Phi_m(u) + \frac{j\pi}{2} \right). \quad (76)$$

The P_j are real Laurent polynomials explicitly determined by the finite recipe in Section 19. Each H_j is entire, uniformly almost periodic on $\mathbb{R}$, and bounded there with every derivative. In particular,

$$a(n) \sim \log 2 (n-1)! \rho^{-n}. \quad (77)$$

The normalized leading error has positive and negative subsequences of nonzero size at scale $n^{-1/4}$.

Thus a constant-coefficient $1/n$ refinement of the leading equivalent cannot describe this sequence. The error is algebraic and oscillatory. The proof does not infer coefficient errors from radial Mellin residues or assume a Delta-domain for F .

13.1 The first four coefficient polynomials

Write b for a positive mode parameter. Then

$$P_0(b) = 1, \quad (78)$$

$$P_1(b) = \frac{2M^2 - 1}{12} b^3 - \frac{b}{2} - \frac{1}{16b}, \quad (79)$$

$$P_2(b) = \frac{1}{4608b^2} \left[(64M^4 - 64M^2 + 16)b^8 + (-192M^3 - 768M^2 + 192)b^6 + (1680M^2 + 312)b^4 - 432b^2 + 81 \right]. \quad (80)$$

For a readable expression at the next order, put

$$\begin{aligned} A_{12} &= 2560M^6 - 3840M^4 + 1920M^2 - 320, \\ A_{10} &= -23040M^5 - 69120M^4 + 11520M^3 + 46080M^2 - 5760, \\ A_8 &= 356544M^4 + 207360M^3 + 141120M^2 - 7632, \\ A_6 &= -544320M^3 - 2177280M^2 + 129600, \\ A_4 &= 1020600M^2 + 267300. \end{aligned}$$

Then

$$P_3(b) = \frac{A_{12}b^{12} + A_{10}b^{10} + A_8b^8 + A_6b^6 + A_4b^4 + 48600b^2 - 30375}{3317760b^3}. \quad (81)$$

The reproducibility package contains exact symbolic expressions and two implementations of their derivation through P_3 . The all-fixed-order theorem is analytic; these four checked polynomials are its displayed specializations, not the basis for an extrapolation from finite data.

14 Exact arithmetic and the signed divisor weight

Expanding the two logarithms formally in (72) gives

$$F(z) = \sum_{m \geq 1} \frac{h(m)}{m} L(z)^m, \quad h(m) = \sum_{d|m} (-1)^{d-1} = \tau(m) - 2\mathbf{1}_{2|m} \tau(m/2). \quad (82)$$

Indeed the coefficient of L^m is $\sum_{kd=m} (-1)^{d-1}/(kd) = h(m)/m$. If $m = 2^v u$ with u odd, then

$$h(m) = (1 - v)\tau(u). \quad (83)$$

Using $L(z)^m = m! \sum_{n \geq m} \begin{bmatrix} n \\ m \end{bmatrix} z^n / n!$ yields the integer formula

$$a(0) = 0, \quad a(n) = \sum_{m=1}^n \begin{bmatrix} n \\ m \end{bmatrix} (m-1)! h(m) \quad (n \geq 1). \quad (84)$$

The Stirling numbers here are unsigned first-kind numbers. The weight h is signed; a positive-mark interpretation is not being asserted. The eventual positivity below follows from asymptotics, not from termwise positivity in (84).

The independent recurrence

$$q_{n,k} = (n-1)q_{n-1,k} + (k-1)q_{n-1,k-1}, \quad q_{1,1} = 1,$$

with all other out-of-range entries zero, computes $q_{n,k} = \begin{bmatrix} n \\ k \end{bmatrix} (k-1)!$. It supplies exact numerators for the portable diagnostics in Section 22 without floating recurrence truncation.

15 A fixed analytic decoration theorem

Let $B(z) = \sum_{k \geq 1} b_k z^k$, with $b_k \geq 0$, be analytic in $|z| < R$. Suppose $B(\rho) = 1$ for $0 < \rho < R$. Impose the difference-span condition

$$\gcd\{k - \ell : b_k b_\ell > 0\} = 1 \quad (85)$$

and positive variance for the law

$$\Pr(K = k) = b_k \rho^k, \quad \mu = \mathbb{E}K, \quad v = \text{Var } K > 0. \quad (86)$$

Let κ_j denote its cumulants. These exist to all orders since $\rho < R$. The support of K lies in the positive integers. Define

$$F_B(z) = \sum_{m \geq 1} \frac{h(m)}{m} B(z)^m, \quad U_n = \rho^n [z^n] F_B(z), \quad \beta_m = \sqrt{\frac{2\pi^2 m}{\mu}}. \quad (87)$$

The notation β_m agrees with (73) when $B = L$.

Theorem 15.1. *For every fixed decoration B as above and every fixed $J \geq 1$,*

$$nU_n = \log 2 + \sum_{j=0}^{J-1} n^{-1/4-j/2} H_{B,j}(\sqrt{n}) + \mathcal{O}_{B,J}(n^{-1/4-J/2}). \quad (88)$$

Its leading function is

$$H_{B,0}(u) = \sum_{m \geq 1} d_o(m) \left(\frac{8\mu}{m} \right)^{1/4} e^{-\pi^2 m v / \mu^2} \sin(2\beta_m u - \pi/4). \quad (89)$$

At order j , the summand is multiplied by a real Laurent polynomial $P_{B,j}(\beta_m)$ and its phase is advanced by $j\pi/2$. In particular,

$$P_{B,1}(b) = \frac{2\mu\kappa_3 - 3v^2}{12\mu^2} b^3 - \frac{v}{2\mu} b - \frac{1}{16b}. \quad (90)$$

All these functions are entire, uniformly almost periodic on the real line, and bounded with all real derivatives.

There is no asserted uniformity as B varies, v tends to zero, the support span changes, or J grows. The difference span is important: $\gcd\{k : b_k > 0\} = 1$ alone does not exclude other points where $|B(\rho e^{i\theta})| = 1$.

16 Smooth renewal interpolation

Let S_k be a sum of k independent copies of K . Coefficient extraction gives the exact finite sum

$$U_n = \sum_{k \geq 1} \frac{h(k)}{k} \Pr(S_k = n). \quad (91)$$

It is finite because $K \geq 1$. Set $\phi(\theta) = B(\rho e^{i\theta})$ and choose its local logarithm $\psi(\theta) = \log \phi(\theta)$ near zero, so

$$\psi(\theta) = \sum_{r \geq 1} \kappa_r (i\theta)^r / r!.$$

Choose an even smooth cutoff η , supported in a sufficiently small symmetric interval $(-\delta, \delta)$ and equal to one near zero. Choose $\chi \in C_c^\infty((c, C))$ for positive c, C , equal to one on a neighborhood of $1/\mu$. Define

$$f_n(x) = \frac{\chi(x/n)}{2\pi x} \int_{\mathbb{R}} \eta(\theta) e^{x\psi(\theta) - in\theta} d\theta. \quad (92)$$

It is real and smooth, and has compact support away from zero.

Lemma 16.1. *For every $A > 0$,*

$$U_n = \sum_{k \geq 1} h(k) f_n(k) + \mathcal{O}_A(n^{-A}). \quad (93)$$

Put $s = n^{-1/2}$, $x = n/\mu + \sqrt{n}y$, and $g_s(y) = n^{3/2} f_n(n/\mu + \sqrt{n}y)$. The family g_s extends smoothly through $s = 0$ in the Schwartz space $\mathcal{S}(\mathbb{R})$. In particular, for every fixed N, p, q ,

$$\sup_y (1 + |y|)^p \left| \partial_y^q \left(g_s(y) - \sum_{j=0}^{N-1} s^j g_j(y) \right) \right| \leq C_{N,p,q} |s|^N, \quad g_j = \frac{\partial_s^j g_s|_{s=0}}{j!}. \quad (94)$$

The leading density is

$$g_0(y) = \frac{\mu^{3/2}}{\sqrt{2\pi v}} e^{-\mu^3 y^2 / (2v)}. \quad (95)$$

Proof. We first establish the smooth family, including continuity in its seminorms. Let

$$t = 1/\mu + sy, \quad a(t) = \chi(t)/t, \quad Q(\theta) = \frac{\psi(\theta) - i\mu\theta}{\theta^2}, \quad Q(0) = -v/2.$$

Extend a smoothly by zero off its positive compact support. Changing variables $\theta = su$ in (92) gives, for $s > 0$,

$$g_s(y) = \frac{1}{2\pi} \int_{\mathbb{R}} a(1/\mu + sy) \eta(su) e^{i\mu y u} \exp\{(1/\mu + sy)u^2 Q(su)\} du. \quad (96)$$

This formula also defines the family for negative s and at zero; it has removed the apparent negative powers of s . The function Q is analytic at zero. Shrink δ until $\operatorname{Re} Q \leq -c_0 < 0$ on the support of η . On the support of a , t stays in a fixed compact subset of $(0, \infty)$. The last exponential is therefore bounded by $e^{-c_1 u^2}$, uniformly in s, y .

For any fixed number of s and y derivatives, the differentiated non-Fourier amplitude is bounded by a polynomial in $|y|, |u|$ times a uniform Gaussian. Before s differentiation, its y dependence occurs through the bounded variable t and through factors of s . Each fixed s derivative introduces at most one additional unabsorbed factor of y . Thus the degree in y is bounded independently of how many later u derivatives are taken. Derivatives of $\eta(su)$ are supported where $|u| \asymp |s|^{-1}$ and are additionally Gaussian-small. Arbitrarily many integrations by parts against $e^{i\mu y u}$ now give uniform bounds for every weighted y derivative, also after any fixed number of s derivatives. Boundary terms vanish by the cutoff and Gaussian decay.

These are not merely bounded-seminorm statements. On each compact y interval, dominated convergence gives continuity of every s, y derivative at $s = 0$. Uniform bounds with a strictly stronger weight control the complementary y tails. Combining the two regions proves continuity in every Schwartz seminorm. The same argument applies to all derivatives; Taylor's theorem in each seminorm gives (94). Substituting $s = 0$ in (96) gives (95). The cutoffs are constant near their central values and disappear from every g_j .

We next compare integer values with (91). Exponential moments exist near zero. Away from a fixed neighborhood of $k/n = 1/\mu$, Chernoff bounds are exponential uniformly in the relevant ratios. For very small k/n , use a fixed positive exponential tilt; for very large k/n , use a fixed negative tilt; on the remaining compact ratio sets use strict convexity. For $k > n$ the probabilities are zero. Inside the central neighborhood, (85) gives $|\phi(\theta)| \leq q < 1$ on the compact complement of the arc where $\eta = 1$. Also $k \asymp n$, so the omitted Fourier arcs cost $\mathcal{O}(q^{c_2 n})$. Multiplication by $|h(k)| \leq \tau(k) \leq k$ and summation over polynomially many indices are harmless. In real-variable cutoff transition regions, $|y|$ is a fixed positive multiple of $\sqrt{n}$; the proved Schwartz bounds make f_n superpolynomially small there. This proves (93). $\square$

Remark 16.2. The interpolation error is only claimed to be superpolynomial. A smooth compact cutoff is not analytic; no contour shift of that cutoff and no exponential global interpolation remainder have been assumed.

16.1 Normalization and the first density correction

Lemma 16.3. *For every $A > 0$,*

$$\int_0^\infty f_n(x) dx = \frac{1}{n} + \mathcal{O}_A(n^{-A}). \quad (97)$$

Consequently $\int g_0 = 1$ and $\int g_j = 0$ for $j \geq 1$.

Proof. Poisson summation applied to the uniformly Schwartz family g_s , with mesh $n^{-1/2}$, makes the difference between the integral and integer sum of f_n smaller than every power of n^{-1} . The unweighted version of Lemma 16.1 identifies that sum with $\rho^n[z^n](-\log(1 - B(z)))$ to the same accuracy. Positivity and the span hypothesis imply that $B(z) = 1$ has no other solution in $|z| \leq \rho$. The zero at ρ is simple since $B'(\rho) > 0$. Hence $(1 - B(z))/(1 - z/\rho)$ is analytic and nonzero on a slightly larger disk, and has an analytic logarithm there. Its coefficients are exponentially small after multiplication by ρ^n . The remaining term $-\log(1 - z/\rho)$ has coefficient $1/n$ under this normalization. The assertions about g_j follow by integrating (94) to arbitrary order. $\square$

Direct differentiation at $s = 0$ also gives

$$\frac{g_1(y)}{g_0(y)} = \left(-\frac{3\mu}{2} + \frac{\kappa_3\mu^2}{2v^2}\right)y + \left(\frac{\mu^4}{2v} - \frac{\kappa_3\mu^5}{6v^3}\right)y^3. \quad (98)$$

It will reproduce the universal first correction (90).

17 The exact signed Voronoi identity

For smooth f of compact support in $(0, \infty)$, the classical divisor Voronoi formula states

$$\begin{aligned} \sum_{k \geq 1} \tau(k) f(k) &= \int_0^\infty f(x) (\log x + 2\gamma) dx \\ &\quad + \sum_{m \geq 1} \tau(m) \int_0^\infty f(x) \mathcal{K}(4\pi\sqrt{mx}) dx, \\ \mathcal{K}(q) &= 4K_0(q) - 2\pi Y_0(q). \end{aligned} \quad (99)$$

The usual boundary term $f(0)/4$ vanishes. We use precisely the compact-support version quoted as Theorem 1.1, equation (8), by Egger–Steiner and credited there to Hejhal [9]. Absolute convergence here follows by integrations by parts for the oscillatory kernel, with the K_0 part exponentially decreasing.

Apply (99) to $f(x)$ and to $2f(2x)$ and subtract. The main integrals differ by $\log 2 \int f$. In the dual part, use the common scale $4\pi\sqrt{mx/2}$. Its coefficient is

$$\mathbf{1}_{2|m} \tau(m/2) - \tau(m) = -d_o(m).$$

Thus the exact signed identity is

$$\sum_{k \geq 1} h(k) f(k) = \log 2 \int_0^\infty f(x) dx - \sum_{m \geq 1} d_o(m) \int_0^\infty f(x) \mathcal{K}(4\pi\sqrt{mx/2}) dx. \quad (100)$$

In particular the oscillatory contribution has sign *positive* $2\pi Y_0$. No estimate for the unsmoothed divisor problem is used.

Apply (100) to f_n . Lemmas 16.1 and 16.3 supply the main term $\log 2$ in nU_n . The support has $cn \leq x \leq Cn$. The standard large-positive-argument bound for K_0 [11] and $d_o(m) \leq m$ make its total contribution $\mathcal{O}(e^{-c_3\sqrt{n}})$, after adjusting the positive constant c_3 . The remaining expression is

$$\sum_{m \geq 1} d_o(m) \int_{\mathbb{R}} g_s(y) 2\pi Y_0 \left(\frac{2\beta_m}{s} \sqrt{1 + \mu sy} \right) dy, \quad s = n^{-1/2}. \quad (101)$$

The all-harmonic justification of its expansion is the next step.

18 A uniform all harmonic remainder

Define the exact normalized Hankel amplitude for $q > 0$ by

$$\mathcal{A}(q) = \sqrt{\frac{\pi q}{2}} e^{-i(q-\pi/4)} H_0^{(1)}(q).$$

The standard coefficients satisfy

$$a_0 = 1, \quad a_k = -\frac{(2k-1)^2}{8k} a_{k-1} \quad (k \geq 1). \quad (102)$$

For every fixed $K, r \geq 0$ (with $K \geq 1$),

$$\partial_q^r \left(\mathcal{A}(q) - \sum_{k=0}^{K-1} i^k a_k q^{-k} \right) = \mathcal{O}_{K,r}(q^{-K-r}). \quad (103)$$

The extra q^{-r} is essential. It follows from the complex Hankel remainder bounds in DLMF 10.17.13–15 [10], uniformly in a small cone around the positive axis, by Cauchy's derivative estimates on disks of radius proportional to q . Merely differentiating a formal real-axis asymptotic expression would not justify this step.

For $\beta = \beta_m$, write

$$\begin{aligned} & \int_{\mathbb{R}} g_s(y) 2\pi Y_0 \left(\frac{2\beta}{s} \sqrt{1 + \mu s y} \right) dy \\ &= 2\sqrt{\pi} \beta^{-1/2} s^{1/2} \operatorname{Im} \left\{ e^{i(2\beta/s - \pi/4)} J_\beta(s) \right\}, \end{aligned} \quad (104)$$

$$J_\beta(s) = \int_{\mathbb{R}} g_s(y) (1 + \mu s y)^{-1/4} e^{i\Psi_{\beta,s}(y)} \mathcal{A} \left(\frac{2\beta}{s} \sqrt{1 + \mu s y} \right) dy, \quad (105)$$

$$\Psi_{\beta,s}(y) = \frac{2\beta\mu y}{\sqrt{1 + \mu s y} + 1}. \quad (106)$$

Every amplitude is understood only on the support of g_s and is smoothly zero-extended with its cutoff. There $1 + \mu s y$ stays in a fixed positive compact interval. The rationalized phase (106) has no singularity at $s = 0$, and

$$\partial_y \Psi_{\beta,s}(y) = \frac{\beta\mu}{\sqrt{1 + \mu s y}} \asymp \beta. \quad (107)$$

Lemma 18.1. *For every fixed $N, L \geq 0$ there is a constant independent of $\beta \geq \beta_1$ and sufficiently small $s \geq 0$ such that*

$$|\partial_s^N J_\beta(s)| \leq C_{N,L} (1 + \beta)^{-L}. \quad (108)$$

Consequently, for every fixed J, L ,

$$J_\beta(s) = \sum_{j=0}^{J-1} C_j(\beta) s^j + R_{J,\beta}(s), \quad |R_{J,\beta}(s)| \leq C_{J,L} s^J (1 + \beta)^{-L}. \quad (109)$$

Proof. The seminorm bounds in (94) permit arbitrary fixed s, y derivatives of the renewal amplitude. For the composed Hankel symbol, use the reciprocal argument

$$p = \frac{s}{2\beta\sqrt{1 + \mu s y}}.$$

Its fixed derivatives are bounded by β^{-1} times polynomials in $|y|$ on the support. The symbol bounds (103), taken to sufficiently high order, give a smooth extension of the composition to $s = 0$, with uniform differentiated bounds. This addresses the apparent singularity of the original argument $q \asymp \beta/s$.

Each s derivative of $e^{i\Psi}$ contributes at most one additional factor of β times a polynomial in y ; more explicitly, $|\partial_s^j \Psi| \leq C_j \beta |y|^{j+1}$. After N derivatives, factor out at most β^N . What remains is a uniformly Schwartz amplitude, including after every fixed number of y derivatives. Now integrate by parts R times using

$$e^{i\Psi} = \frac{1}{i\partial_y \Psi} \partial_y (e^{i\Psi}).$$

The reciprocal gradient is $\sqrt{1 + \mu s y}/(\beta \mu)$. It and its required y derivatives supply β^{-1} per integration; the other factors are bounded powers of $1 + \mu s y$ and powers of s . All boundary terms vanish. The resulting absolute integral is at most $C_{N,R} \beta^N$. Choose $R > N + L$ to obtain (108). The smooth extension and seminorm continuity give the same bound at $s = 0$. Taylor's theorem with its integral remainder proves (109). $\square$

18.1 Summing the actual remainders

The prefactor in (104) is $2\sqrt{\pi} \beta_m^{-1/2} s^{1/2}$. Since $d_o(m) \leq m$ and $\beta_m \asymp \sqrt{m}$, the sum of remainder bounds is at most

$$C s^{J+1/2} \sum_{m \geq 1} m^{3/4-L/2} = \mathcal{O}(s^{J+1/2}) \quad (L > 4).$$

This is $\mathcal{O}(n^{-1/4-J/2})$. It bounds the actual remainder over *all* harmonics. There is no step that sums only fixed-mode asymptotic error statements whose constants could grow with m .

The coefficients C_j are finite Gaussian Fourier moments, so

$$C_j(\beta) = i^j e^{-v\beta^2/(2\mu)} P_{B,j}(\beta) \tag{110}$$

with real Laurent polynomials $P_{B,j}$. For $j = 0$, the Gaussian transform in (95) is $e^{-v\beta^2/(2\mu)}$. Multiplying it by $2\sqrt{\pi} \beta_m^{-1/2}$ gives

$$\left(\frac{8\mu}{m}\right)^{1/4} e^{-\pi^2 m v / \mu^2},$$

which verifies all constants in (89). The higher coefficients follow from the recipe below, and (98) gives (90). This proves Theorem 15.1. For each fixed j , the exponential damping in m dominates every polynomial factor, every real derivative, and $e^{C\sqrt{m}|\operatorname{Im} u|}$ on compact complex u sets. Thus the mode series is entire and its derivatives converge uniformly on $\mathbb{R}$. Uniform approximation by finite trigonometric sums proves uniform almost periodicity.

19 A finite recipe for every fixed order

This section makes (110) constructive and connects the proof to the exact symbolic programs. All series in s are truncated at the finite order needed; no convergence as $J \rightarrow \infty$ is asserted. Let $\sigma^2 = v/\mu^3$, with g_0 as in (95), and define

$$E_j(y, u) = \frac{\kappa_{j+2}(iu)^{j+2}}{\mu(j+2)!} + \frac{y\kappa_{j+1}(iu)^{j+1}}{(j+1)!}. \tag{111}$$

Expand

$$(1 + \mu sy)^{-1} \exp \left(\sum_{j \geq 1} E_j(y, u) s^j \right).$$

In the coefficient of each s^j , replace $(iu)^r$ by $\mu^{-r} g_0^{(r)}(y)/g_0(y)$. The result is $G_j(y) = g_j(y)/g_0(y)$. This is exactly Gaussian differentiation under the Fourier integral in (96).

Next set

$$R(s, y) = \sum_{r \geq 1} 2\beta \binom{1/2}{r+1} \mu^{r+1} y^{r+1} s^r, \quad (112)$$

$$T(s, y) = \left(\sum_{j \geq 0} G_j(y) s^j \right) (1 + \mu sy)^{-1/4} e^{iR(s, y)} \cdot \sum_{k \geq 0} a_k \left(\frac{is}{2\beta} \right)^k (1 + \mu sy)^{-k/2}. \quad (113)$$

The coefficients are

$$C_j(\beta) = \int_{\mathbb{R}} g_0(y) e^{i\mu\beta y} [s^j] T(s, y) dy, \quad (114)$$

$$P_{B,j}(\beta) = i^{-j} e^{v\beta^2/(2\mu)} C_j(\beta). \quad (115)$$

Only finitely many cumulants, powers and Gaussian moments are needed at any fixed order. The parity $G_j(-y) = (-1)^j G_j(y)$ and the phase and symbol parities make $P_{B,j}$ real. These formulas explain the successive half-powers in the coefficient expansion.

19.1 Specialization to the logarithmic decoration

For $B = L$, the cumulant generating function is

$$\log B(\rho e^t) = \log \left(1 - \log(1 - M(e^t - 1)) \right). \quad (116)$$

In particular,

$$\begin{aligned} \kappa_1 &= \kappa_2 = M, \\ \kappa_3 &= M(M^2 + 1), \\ \kappa_4 &= M(M^3 + 6M^2 + 1), \\ \kappa_5 &= M(8M^4 + 10M^3 + 25M^2 + 1). \end{aligned}$$

Put $z = My$. The base measure becomes the standard Gaussian $\gamma(z) dz = (2\pi)^{-1/2} e^{-z^2/2} dz$. The Gaussian-derivative replacement is $(iu)^r \mapsto (-1)^r \text{He}_r(z)$ in these scaled coordinates, where He_r is the probabilists' Hermite polynomial. The final normalized Fourier moment is $z^r \mapsto i^r \text{He}_r(\beta)$. The script `derive_coefficients.py` implements these exact rules and produces (78)–(81).

A second script, `verify_coefficients.py`, uses actual symbolic Gaussian derivatives rather than a Hermite routine, and the integration-by-parts moment recurrence

$$m_0 = 1, \quad m_1 = ib, \quad m_{r+1} = ib m_r + r m_{r-1}. \quad (117)$$

It reconstructs $G_0, \dots, G_3$ and the first four kernel coefficients before comparing with the saved polynomials. For transparency, write $D_r = \gamma^{(r)}/\gamma$ and $c_j = \kappa_j/M$. The third density coefficient is

$$G_3 = -z^3 + \frac{z^3 D_2}{2} - \frac{z^3 D_4}{8} + \frac{z^3 D_6}{48} + \frac{c_5 D_5}{120} \\ + \left(\frac{c_3^2}{72} + \frac{c_4}{48} \right) z D_6 + \left(\frac{c_3 c_4}{144} + \frac{c_3 z^2}{48} \right) D_7 + \frac{c_3^2 z D_8}{144} + \frac{c_3^3 D_9}{1296}. \quad (118)$$

The cubic kernel coefficient, after multiplying the geometric, phase and Hankel factors, is

$$Q_3 = -\frac{25z^3}{256} + \frac{5b^2 z^5}{128} + \frac{45z}{2048b^2} \\ + i \left(\frac{b^3 z^6}{384} - \frac{75b z^4}{512} - \frac{75z^2}{2048b} + \frac{75}{8192b^3} \right). \quad (119)$$

If Q_1, Q_2 denote the lower kernel coefficients, applying (117) to $G_3 + G_2 Q_1 + G_1 Q_2 + Q_3$ and dividing by i^3 gives (81). The checks use exact symbolic identities, not a numerical fit.

20 Nonvanishing and signed subsequences

For the logarithmic decoration the first harmonic dominates. From $d_o(m) \leq m$ and $m^{-1/4} \leq 1$,

$$\sum_{m \geq 2} \frac{K_m}{K_1} \leq \sum_{m \geq 2} m r^{m-1} = \frac{1}{(1-r)^2} - 1 < \frac{1}{100}. \quad (120)$$

Here is an exact rational certificate for the last step. The elementary bounds $e < 87/32$ and $\pi > 157/50$ imply

$$\frac{\pi^2}{e-1} > \frac{(157/50)^2}{55/32} > \frac{573}{100}.$$

The positive Taylor sum $\sum_{k=0}^{20} (573/100)^k/k! > 300$ proves $r < 1/300$. Then $1/(1-r)^2 - 1 < 599/89401 < 1/100$. The checker verifies these rational inequalities and an exponential-series upper bound on e ; it takes the elementary bound $\pi > 157/50$ as an input.

Choose $u_\ell \rightarrow \infty$ with $\Phi_1(u_\ell) = \pi/2 + 2\pi\ell$, and let n_ℓ be a nearest integer to u_ℓ^2 . Then $\sqrt{n_\ell} - u_\ell = \mathcal{O}(1/u_\ell)$. Bounded derivatives and (120) give $H_0(\sqrt{n_\ell}) \geq 0.99K_1 + o(1)$. Choosing phase $3\pi/2 + 2\pi\ell$ instead gives $H_0(\sqrt{n_\ell}) \leq -0.99K_1 + o(1)$. By (75),

$$n^{1/4} \left(\frac{n \rho^n a(n)}{n!} - \log 2 \right)$$

has positive and negative subsequences bounded away from zero. In particular, the error is not $o(n^{-1/4})$ and cannot be $\mathcal{O}(n^{-1})$. These conclusions do not identify an exact limsup or liminf, a limiting distribution, or independence of the harmonic phases.

21 Eventual shape and smooth inverse envelopes

Corollary 21.1. *The sequence $a(n)$ is eventually positive, strictly increasing, and strictly log-convex. More precisely,*

$$\frac{a(n+1)}{a(n)} \sim \frac{n}{\rho}, \quad \log \frac{a(n-1)a(n+1)}{a(n)^2} = \frac{1}{n} + \mathcal{O}(n^{-5/4}). \quad (121)$$

No explicit onset is asserted.

Proof. Positivity and the ratio follow immediately from (77). For fixed $J \geq 1$, define the eventually positive smooth function

$$A_J(t) = \Gamma(t)\rho^{-t} \left[\log 2 + \sum_{j=0}^{J-1} t^{-1/4-j/2} H_j(\sqrt{t}) \right]. \quad (122)$$

With $J = 2$, (75) implies $\log a(n) = \log A_2(n) + \mathcal{O}(n^{-5/4})$. Bounded derivatives of the H_j and the usual digamma estimates give $(\log A_2)''(t) = 1/t + \mathcal{O}(t^{-5/4})$. For any twice differentiable f ,

$$f(n+1) - 2f(n) + f(n-1) = \int_{-1}^1 (1-|u|)f''(n+u) du.$$

Apply this identity to $\log A_2$ and bound the three discrete $\mathcal{O}(n^{-5/4})$ errors separately. This proves the second assertion in (121), eventually positive. No discrete error term has been differentiated. $\square$

For fixed J , logarithmic differentiation of (122) gives

$$(\log A_J)'(t) = \log(t/\rho) + \mathcal{O}(t^{-1}) + \mathcal{O}_J(t^{-3/4}) > 0 \quad (123)$$

for sufficiently large t . Also, with $\alpha = 1/4 + J/2$,

$$a(n) = A_J(n)(1 + \mathcal{O}_J(n^{-\alpha})). \quad (124)$$

Fix an integer n_0 beyond the eventual positivity and monotonicity onset, and define

$$N(x) = \min\{n \geq n_0 : a(n) \geq x\}. \quad (125)$$

The number n_0 is fixed, not allowed to vary with x . Let $t_J(x)$ be the unique sufficiently large solution of $A_J(t) = x$.

Theorem 21.2. *There is a constant C_J such that, for every sufficiently large x ,*

$$\left\lceil t_J - C_J \frac{t_J^{-\alpha}}{\log t_J} \right\rceil \leq N(x) \leq \left\lceil t_J + C_J \frac{t_J^{-\alpha}}{\log t_J} \right\rceil, \quad \alpha = \frac{1}{4} + \frac{J}{2}. \quad (126)$$

The displacement shrinks to zero. The constants and onset are existential. This is not an unconditional single-ceiling rule at jumps.

Proof. From (124), $|\log a(n) - \log A_J(n)| \leq Cn^{-\alpha}$ for large integers. Put $t = t_J$ and $\delta = C't^{-\alpha}/\log t$. For C' sufficiently large, (123) makes the decrease of $\log A_J$ from t to any nearby integer below $t - \delta$ dominate this pointwise error. Those integers have $a(n) < x$. Similarly the integer $\lceil t + \delta \rceil$ has $a(n) \geq x$. The derivative bounds are uniform throughout a fixed neighborhood of t , and enlarging C' handles variation across one lattice spacing. Eventual monotonicity handles earlier integers. This proves both ceilings, even when x lies arbitrarily close to a jump. $\square$

Any prescribed finite prefix has a finite maximum. Once x exceeds that maximum, it cannot change the threshold; thus (125) agrees eventually with the full-sequence convention if that convention is used. No positivity or monotonicity of the prefix is assumed.

[write] **7 October 2026.** Theorem 21.2 brackets the threshold directly at the integers and is not an instance of the transseries volume's `p0:thm:staircase`; the Lambert seed (128) is an instance of its `p0:prop:factorial-core`, and (127) is an analogue of the first term of its operator series (front matter, “The inverses and the transseries volume”).

21.1 A useful first inverse correction

Let $b = b(x)$ solve $\Gamma(b)\rho^{-b}\log 2 = x$ on its increasing large branch, and put $D(b) = \psi_0(b) - \log \rho$, where ψ_0 is the digamma function. For every fixed $J \geq 1$,

$$t_J(x) = b - \frac{b^{-1/4}H_0(\sqrt{b})}{\log 2 D(b)} + \mathcal{O}\left(\frac{b^{-1/2}}{\log b}\right). \quad (127)$$

This follows by logarithmic Taylor expansion and (123). The square in $\log(1 + \varepsilon)$ contributes at order $b^{-1/2}/\log b$; retaining only the next oscillatory H_1 term would miss it. The integer inverse still requires (126). A Lambert- W starting approximation from Stirling's formula is

$$Y = \log \frac{x}{\sqrt{2\pi \log 2}}, \quad b = \frac{Y}{W(Y/(e\rho))} + \mathcal{O}(1), \quad (128)$$

where the positive real branch is used for large x . Ordinary monotone scalar solving of (122) preserves all the displayed oscillations. No new inversion principle is claimed.

22 Finite checks and portable reproduction

The public package uses exact integer arithmetic for the sequence and arbitrary-precision scalar arithmetic for the diagnostics. It does not rely on the platform-dependent precision of a C or NumPy `longdouble`, and it contains no truncated floating Stirling recurrence. Finite checks support transcription and implementation; they do not prove an asymptotic remainder bound or a shape onset.

22.1 Exact checks

The supplied OEIS b-file has exactly 401 entries, $n = 0, \dots, 400$, and is included only as the numerical fixture needed for reproduction, with attribution to Václav Kotesovec and OEIS [8]. Every entry agrees with (84). A separate exact rational expansion of the nested logarithms agrees through degree 24. The checker also verifies the signed divisor identities through 400 and the rational first-harmonic certificate. All checks use explicit exceptions, so Python optimization does not disable them.

The coefficient generator and the independent Gaussian-differentiation check both reproduce P_0 through P_3 exactly. The latter also reconstructs G_0 through G_3 and cumulants through order five. Only these four displayed correction polynomials enter the public numeric results.

22.2 Diagnostic values

Write $V_n = n\rho^n a(n)/n!$ and define

$$E_J(n) = n^{1/4+J/2} \left(V_n - \log 2 - \sum_{j=0}^{J-1} n^{-1/4-j/2} H_j(\sqrt{n}) \right). \quad (129)$$

Here $J = 0$ denotes an empty sum; $J = 4$ means all four polynomials $P_0, \dots, P_3$ have been used. Table 4 is generated from exact integer numerators through 2000, evaluated at 80 decimal digits with 40 harmonics. Repeating at 100 digits and 60 harmonics changes every stored scalar by less than 10^{-68} . That is an observed numerical stability check, not a rigorous interval bound on the

n	V_n	$E_1(n)$	$E_2(n)$	$E_3(n)$	$E_4(n)$
100	0.690920238401	-0.0177350	0.2700278	0.0189424	0.4122755
200	0.695137609459	0.0412967	-0.2399859	0.0340109	-0.1242122
400	0.692560796792	-0.0890456	-0.0965587	0.1251426	0.0796437
800	0.693393950282	-0.0741678	-0.2093545	-0.0459881	-0.2464049
1500	0.692148459685	-0.0394972	0.2728415	-0.0436450	0.2898864
2000	0.693987426187	0.0578197	-0.2356268	-0.1050114	-0.0802815

Table 4: Finite 80-digit diagnostics, rounded for display. Each E_J uses J oscillatory terms and the scale in (129).

infinite harmonic tails or on the asymptotic remainder. Neither bounded-looking scaled errors nor agreement at these indices certifies the existence constant in (75). In particular, the table does not provide numerical values for C_J or n_0 .

At exact levels $x = a(n)$, a smooth inverse can lie on either side of n . Table 5 records the computed offsets for two truncations. A positive offset makes a direct single ceiling return $n + 1$ at that level; this illustrates the need for the two-envelope formulation rather than establishing any effective envelope constant.

n	$t_1(a(n)) - n$	$t_4(a(n)) - n$
100	-1.604049×10^{-4}	3.729361×10^{-6}
400	-2.227586×10^{-4}	2.492283×10^{-8}
2000	3.457080×10^{-5}	$-5.365932 \times 10^{-10}$

Table 5: Diagnostic smooth-inverse offsets at exact integer sequence levels. These are numerical observations, not certified interval enclosures.

22.3 Commands and integrity

The package README gives ordinary Python and `python -O` commands. Its verifier checks a pinned manifest with explicit guards, replays the exact and diagnostic programs, compares saved outputs, and checks the TeX table fields against the generated JSON. A separate ZIP verifier checks its actual member set, every member byte and the whole-ZIP SHA256 against a release receipt. The manifest excludes only itself to avoid self-reference. No third-party paper full texts, private review notes, machine-specific paths, or credentials are included. Rebuilding the PDF requires a standard pdfLaTeX installation; exact PDF-byte reproduction is checked in the stated build environment, while mathematical reproduction does not depend on PDF byte identity.

[write] 7 October 2026. In this report the package’s programs are `code/211-harmonic-*.py` (those of `repro/` as `code/211-harmonic-repro-*.py`) and its fixture and outputs `data/211-harmonic-repro-*`. Not shipped: `Report211.tex` (its text is this Part), the delivered PDF, the package `README.md` and `MANIFEST.json`. The release receipt `Report211-repro-v1.receipt.json` that the ZIP verifier needs was not delivered (only the ZIP arrived), so `verify_zip.py` cannot be run as the package documents; `verify_package.py` needs the manifest and the delivered layout. The report’s README gives routes that run the exact and coefficient checks on a copy.

23 Sources and bounded comparison

OEIS evidence. The inspected A330498 export is revision 13, dated 17 December 2019; it supplies the defining EGF and the conjectural $\log 2$ label. The b-file was retrieved from its official OEIS URL on 4 October 2026. The record contains no linked proof or higher correction in the inspected material. We do not infer worldwide absence from this fact. The independent report-writing attempt to reopen the live entry returned a retrieval error, so the wording here remains tied to the inspected saved export rather than claiming a newly observed live revision.

[write] 7 October 2026. The write reopened the live entry on 7 October 2026: it is revision #13, the revision inspected here, and it still labels $c = \log 2$ a conjecture.

Classical mathematical inputs. The compact-support divisor formula is explicitly available in Egger–Steiner, Theorem 1.1, equation (8), credited there to Hejhal [9]. The full Egger–Steiner PDF and its theorem were inspected. The attempted original Hejhal full text was behind a subscription page; its full text was not read. We therefore cite the compact-support theorem as actually available in Egger–Steiner, rather than claiming to have inspected the original proof. Their later Laplace-transform proof concerns another function class and is not silently substituted for that compact-support statement. The Hankel coefficients and complex remainder bounds come from DLMF 10.17, and the decaying modified-Bessel estimate from DLMF 10.40 [10, 11]. The renewal interpolation is proved here using classical Fourier local-limit and Edgeworth ideas.

Relation to Report185. The earlier Report185 studies $\sum_{k \geq 1} \log(1 + B^k)$, without the factor $1/k$, and explicitly excludes A330498. Its arithmetic weight is $\sigma(\text{odd}(m))$, its exact transformation is an eta identity, and its coefficient main term is a pole with normalized first correction $n^{-3/4}$. Here the harmonic factor produces $h(m)$, a cancelled logarithmic divisor mean, a signed Voronoi kernel, and a relative first correction $n^{-1/4}$. Shared analytic-decoration hypotheses and variance damping are acknowledged. This is a distinct kernel continuation, not a claim to invent Voronoi asymptotics, Gaussian smoothing, or inversion.

[write] 7 October 2026. Report185 is Part I of this report. Its distinctness from this Part is as stated here; the front matter (“Guide to this report”) adds the write’s observation that the two leading oscillations have the same frequencies, damping and phase and differ only in their amplitudes.

Other bounded comparisons. The inspected ProveIt report descriptions for A301746/A294363 concern divisor-square partition products and exponential divisor assemblies; the A124380 report concerns a different signed-moment integral with an exponentially smaller oscillatory sector [12, 13]. The inspected formulas do not establish the harmonic composition in this report. Exact-ID and semantic searches are limited evidence, and no exhaustive repository or worldwide literature theorem is inferred from their negative results.

Claims left outside scope. This report does not classify all singularities of F_B , assert a Delta-domain, give an optimally truncated or exponentially improved expansion, prove a variance-zero transition uniform in B , or supply effective remainder constants. It does not certify a numerical positivity or log-convexity onset, a single-ceiling inverse formula at every jump, an exact oscillation distribution, or global priority. The contribution proved here is the target-specific constant and the fixed-decoration, all-fixed-order oscillatory transfer with a summable remainder.

Further calculations. A natural next calculation would make selected tail and interpolation constants explicit, with a verified finite range preceding the asymptotic regime. That would require quantitative cutoff choices, Hankel remainder estimates, and certified evaluation of the retained harmonics, rather than fitting the diagnostic table. A separate extension could study how the damping and relevant harmonics change as a specified family of decorations approaches zero variance. The present fixed-decoration theorem supplies neither uniform transition bounds nor effective constants; these are proposed extensions of this report, not claims about their status throughout the literature.

[write] **7 October 2026.** The claims left outside scope and these further calculations are gathered with Part I’s questions in Section 24.

24 Further questions and research

[Added 7 October 2026, batch 110.] This section gathers, under Vladimir’s standing rule of 4 October 2026, every question and every claim left unproved by the two manuscripts: Part I’s Section 12, the closing remarks of its Sections 7 and 8, and Part II’s paragraphs “Claims left outside scope” and “Further calculations” (Section 23). The delivered texts stay where they are; each item names its source, the argument or evidence offered, and what is missing. No claim of either manuscript was found to be false.

1. *Effective constants, cutoffs and onsets* (Part I, item 1; Part II, “Claims left outside scope” and “Further calculations”). All remainder constants and starting indices in Theorems 1.1, 3.1, 9.1, 13.1, 15.1 and 21.2 and in Corollary 21.1 are existence constants. Both proofs track polynomial-times-exponential bounds; what is missing is an explicit choice of the cutoffs, explicit stationary-phase (Part I) and Hankel and interpolation (Part II) remainder constants, and a certified evaluation of the retained harmonics, together with a verified finite range below the asymptotic regime. With them the two-ceiling envelopes would become certified inverse algorithms, and the onsets of positivity, increase and log-convexity of A330498 would become numbers. The diagnostic tables (Tables 2, 3, 4, 5) supply none of these constants.
2. *Other support spans* (Part I, item 2; Part II, after Theorem 15.1). If the difference span exceeds one, $|B(\rho e^{i\theta})| = 1$ at several points and each contact contributes its own phases and cusps; the example $(z + z^3)/2$ shows that the condition $\gcd\{j : b_j > 0\} = 1$ does not suffice. Open for both kernels.
3. *Vanishing variance* (Part I, item 3; Part II, “Claims left outside scope” and “Further calculations”). As $v/\mu^2 \rightarrow 0$ the number of appreciable harmonics grows, since the damping is $e^{-\pi^2 mv/\mu^2}$; a crossover theorem uniform in a family of decorations approaching a monomial needs new two-parameter estimates. Neither fixed- B theorem gives it.
4. *Joint phases, extremes and limiting error laws* (Part I, item 4 and the remark after Theorem 7.2; Part II, Section 20 and “Claims left outside scope”). The frequencies are proportional to $\sqrt{m}$; the signed subsequences use only the first harmonic and determine neither the exact lim sup and lim inf of the scaled errors nor a limiting distribution of $H_0(\sqrt{n})$, nor whether the harmonic phases behave independently. Open.
5. *Growing orders, optimal truncation and exponentially improved expansions* (Part I, item 5; Part II, Scope and “Claims left outside scope”). The recipes are finite at every fixed order and say nothing about the growth of the H_j or P_j as $j \rightarrow \infty$, optimal truncation, or a complete transseries. Open.

6. *Singularities and continuation domains* (Part I, the remark after Corollary 8.2; Part II, “Claims left outside scope”). Part I proves that odd-order root-of-unity preimages are genuine cusps and that no standard Delta-domain exists at ρ ; it does not classify all cusps or all continuation domains of F_B . Part II asserts no Delta-domain for its kernel and classifies no singularities; Part I’s cusp argument is not claimed for the harmonic kernel. Open.
7. *Other weights* (Part I, item 6). Part I’s item 6 says that the harmonic weight of A330498 “leads to a different transformation problem” and that other products or weights “should be treated from their own exact kernels”. For the weight $1/k$ this is now done: Part II proves the corresponding all-orders theorem by a Voronoi identity in place of the eta identity. For other weights w_k the question is open; the Guide’s observation that both Parts’ leading modes share frequencies, damping and phase suggests asking which weights keep that structure, but nothing is proved about it.
8. *One ceiling at the jumps* (Part I, Section 9; Part II, Theorem 21.2 and “Claims left outside scope”). Both Parts prove two-ceiling envelopes and state that they cannot be collapsed to one ceiling of a smooth chart at every jump; by the transseries volume’s `p0:thm:staircase(5)` no continuous function does that uniformly. Which thresholds a given chart decides (the separation condition of item (2) there) is a finite question for each Y , not settled by either Part.

Part II also declines a claim of global priority; that is a statement about the literature, not a mathematical question, and it stays as Part II’s non-claim.

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